SEMICONDUCTOR DESIGN

TOP · Physical Implementation

IC Compiler II

Setup & Scenario · Floorplan · Place · CTS · Route · Sign-off

DC netlist(TOP)을 SAED32 · 72 × 72 μm boundary에 배치 · 배선, 2 scenario MCMM으로 sign-off 점검까지.

정태웅 · 청주대학교 · IC Compiler II

Agenda

spec의 IC Compiler II flow 순서대로 진행 — setup부터 sign-off까지.

01

Setup & Scenario

NDM · nlib 구성, routing layer, 1 mode × 2 corner MCMM.

02

Floorplan & Power Plan

72 × 72 die, port constraint 보정, ring · mesh · rail.

03

Place

congestion 예측 후 place_opt → refine_opt.

04

CTS

clock tree build · route, NDR, post-CTS clock 제약.

05

Route

route_auto → route_opt, zroute error 해결.

06

Filler & Sign-off

filler 삽입 후 DRC · LVS · PG · timing 최종 점검.

ICC2 Flow

spec flow 순서대로 단계별 tcl 실행 → check · report로 결과 확인 후 다음 단계 진행.



zroute error → ref library 교체 → 재실행

Library
logical(.db) · physical(LEF) library로 만든 NDM을 ref_libs로 묶어 saed32.nlib 생성.

Scenario
mode normal × corner tt0p78v125c / tt1p05vn40c = scenario 2개, 각 scenario에 MY_CON.con 적용.

TOP Design & Spec

max / min / error pipeline — DESIGN1 (TOP) spec 중 ICC2 관련 항목.

Function
imax1–4 중 최댓값, imin1–4 중 최솟값(4-bit), error = |max – min| 출력.

Pipeline
input regs → mid regs → output regs.

I/O
입력 imax* · imin* · clk · reset, 출력 max_value · min_value · error (각 4-bit).

항목	spec
Logical library	saed32*vt_tt0p78v125c.db(slow) / saed32*vt_tt1p05vn40c .db(fast)
Physical library	saed32nm_*vt_1p9m.lef
Tech	saed32nm_1p9m_mw.tf
TLU+	saed32nm_1p9m_Cmax.tluplus / saed32nm_1p9m_Cmin.tluplus
Boundary	{0 0} {72 72}, shape R
Routing layer	TOP M4 · LOW M1
Ring / Mesh	Vertical M4 · Horizontal M3, ring via VIA34SQ_C {2 2}
Design rule	fanout 12 · cap 10 (spec 2.0 → 완화) · transition 0.12
Vth	LVT < 20 %

01

SECTION

Setup & Scenario

logical · physical library로 만든 NDM을 nlib으로 묶고, PVT 설정과 scenario 작성.

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setup.tcl (1/2)

library 생성 → netlist import → routing layer → MCMM scenario.

setup.tcl · 1–4

```
#-------------------------------------------------------------------------------
# 1. Library creation
#    - technology lib + reference NDMs (LVT / HVT / RVT / physical-only)
#-------------------------------------------------------------------------------
create_lib saed32.nlib \
    -use_technology_lib TECH_LIB.ndm \
    -ref_libs {
        saed32lvt_tt_c.ndm
        saed32hvt_tt_c.ndm
        saed32rvt_tt_c.ndm
        saed32_physical_only.ndm
        TECH_LIB.ndm
    }
save_lib

#-------------------------------------------------------------------------------
# 2. Netlist import
#-------------------------------------------------------------------------------
read_verilog -top TOP ../dc/output/TOP_netlist.v
link_block

#-------------------------------------------------------------------------------
# 3. Routing layer range (signal routing: M1 ~ M4)
#-------------------------------------------------------------------------------
set_ignored_layers -min_routing_layer M1 -max_routing_layer M4
report_ignored_layers

#-------------------------------------------------------------------------------
# 4. MCMM : 1 mode x 2 corners = 2 scenarios
#-------------------------------------------------------------------------------
create_mode   normal
create_corner tt0p78v125c
create_corner tt1p05vn40c
create_scenario -mode normal -corner tt0p78v125c -name normal_tt0p78v125c
create_scenario -mode normal -corner tt1p05vn40c -name normal_tt1p05vn40c

# Apply constraints (SDC) to each scenario
foreach scn {normal_tt0p78v125c normal_tt1p05vn40c} {
    current_scenario $scn
    source ../dc/MY_CON.con
}
```

nlib

Logical library → saed32*vt_tt_c.ndm, saed32_physical_only.ndm
physical library / Tech / TLU+ → TECH_LIB.ndm

Routing layer

set_ignored_layers로 signal routing을 M1 – M4로 제한.

MCMM

mode normal × corner 2 → scenario 2개, 각 scenario에 MY_CON.con source.

setup.tcl (2/2)

corner별 PVT · timing derate · TLU+ 설정.

setup.tcl · 5–7

```
#-------------------------------------------------------------------------------
# 5. corner PVT
#-------------------------------------------------------------------------------
current_corner tt0p78v125c
set_temperature 125
set_process_number 1.00
set_voltage 0.78 -object_list VDD
set_timing_derate -late  1.05 -cell_delay -net_delay
set_timing_derate -early 0.92 -cell_delay -net_delay

current_corner tt1p05vn40c
set_temperature -40
set_process_number 1.00
set_voltage 1.05 -object_list VDD
set_timing_derate -late 1.05 -net_delay -cell_delay
set_timing_derate -early 0.92 -net_delay -cell_delay

#-------------------------------------------------------------------------------
# 6. Parasitic tech (TLU+)
#-------------------------------------------------------------------------------

# Per-corner parasitic parameters (original behavior)
#   tt0p78v125c : Cmax, 125C
#   tt1p05vn40c : Cmin, -40C
set_parasitic_parameters -corners tt0p78v125c \
    -early_spec maxTLU -early_temperature 125 \
    -late_spec  maxTLU -late_temperature  125 -library TECH_LIB
set_parasitic_parameters -corners tt1p05vn40c \
    -early_spec minTLU -early_temperature -40 \
    -late_spec  minTLU -late_temperature  -40 -library TECH_LIB

report_corners
report_parasitic_parameters

#-------------------------------------------------------------------------------
# 7. Save
#-------------------------------------------------------------------------------

report_corners > reports/setup/report_corners_after_setup.txt
report_parasitic_parameters > reports/setup/report_parasitic_parameters_after_setup.txt
report_ignored_layers > reports/setup/report_ignored_layers_after_setup.txt

save_lib -as nlib/design_setup.nlib
```

PVT

tt0p78v125c: 0.78 V · 125 °C
tt1p05vn40c: 1.05 V · −40 °C

Timing derate

두 corner 모두 cell · net late 1.05 / early 0.92.

TLU+

125 °C corner는 Cmax, −40 °C corner는 Cmin.

Routing Layer

report_ignored_layers — signal routing 범위 M1 – M4.

report_ignored_layers

```
****************************************
Report : Ignored Layers
Design : TOP
Version: T-2022.03-SP5
Date   : Thu Oct  8 14:18:29 2026
****************************************
Layer Attribute                    Value
----------------------------------------
Min Routing Layer                  M1
Max Routing Layer                  M4
RC Estimation Ignored Layers       PO M5 M6 M7 M8 M9 MRDL
```

MCMM Scenario

report_scenarios — 1 mode × 2 corner = 2 scenario.

report_scenarios

```
****************************************
Report : scenario
Design : TOP
Version: T-2022.03-SP5
Date   : Thu Oct  8 14:23:24 2026
****************************************

                                                                                                                  Leakage  Dynamic                                  Cell  Signal
Name            Mode            Corner          Active  Setup  Hold  Power    Power    Max_tran  Max_cap  Min_cap  EM    EM
-------------------------------------------------------------------------------------------------------------------------------
normal_tt0p78v125c *
                normal          tt0p78v125c     true    true   true  true     true     true      true     true     false false
normal_tt1p05vn40c *
                normal          tt1p05vn40c     true    true   true  true     true     true      true     true     false false
```

Corner — tt0p78v125c

0.78 V · 125 °C — 저전압 · 고온 corner.

report_corners · tt0p78v125c

```
--------------------------------------------------------------------------------
Corner tt0p78v125c
--------------------------------------------------------------------------------

Current: false  Default: false  Empty: false

Scenarios associated with this corner:

                                                 Leakage   Dynamic
Scenario           Mode           Active  Setup  Hold  Power     Power     Max_tran  Max_cap  Min_cap  Cell_em  Signal_em
-------------------------------------------------------------------------------------------------------------------------
normal_tt0p78v125c  normal         true    true   true  true      true      true      true     true     false    false


Top-Level PVT Settings:
  early process label             (none)
  early process number            1
  early voltage                   0.78
    source                        UPF
    file/line                     --
  early temperature               125.00
    source                        hierarchical setting
    file/line                     --

  late process label              (none)
  late process number             1
  late voltage                    0.78
    source                        UPF
    file/line                     --
  late temperature                125.00
    source                        hierarchical setting
    file/line                     --



0 objects with explicit PVT settings.


Top-level cell timing derates:
 data_cell derate early:   rise: 0.92    fall: 0.92
 clk_cell derate early:   rise: 0.92    fall: 0.92
 cell_check derate early:   rise: 1.00    fall: 1.00
 data_cell derate  late:   rise: 1.05    fall: 1.05
 clk_cell derate  late:   rise: 1.05    fall: 1.05
 cell_check derate  late:   rise: 1.00    fall: 1.00

Global net timing derates:
 net_data, early:   rise: 0.92    fall: 0.92
 net_data, late:   rise: 1.05    fall: 1.05
 net_clock, early:   rise: 0.92    fall: 0.92
 net_clock, late:   rise: 1.05    fall: 1.05
```

PVT
process 1 · 0.78 V · 125 °C (early / late 동일).

Cell derate
data · clk cell early 0.92 / late 1.05, cell_check 1.00.

Net derate
net_data · net_clock early 0.92 / late 1.05.

Corner — tt1p05vn40c

1.05 V · −40 °C — 고전압 · 저온 corner.

report_corners · tt1p05vn40c

```
--------------------------------------------------------------------------------
Corner tt1p05vn40c
--------------------------------------------------------------------------------

Current: true  Default: false  Empty: false

Scenarios associated with this corner:

                                                 Leakage   Dynamic
Scenario           Mode           Active  Setup  Hold  Power     Power     Max_tran  Max_cap  Min_cap  Cell_em  Signal_em
-------------------------------------------------------------------------------------------------------------------------
normal_tt1p05vn40c  normal         true    true   true  true      true      true      true     true     false    false


Top-Level PVT Settings:
  early process label             (none)
  early process number            1
  early voltage                   1.05
    source                        UPF
    file/line                     --
  early temperature               -40.00
    source                        hierarchical setting
    file/line                     --

  late process label              (none)
  late process number             1
  late voltage                    1.05
    source                        UPF
    file/line                     --
  late temperature                -40.00
    source                        hierarchical setting
    file/line                     --



0 objects with explicit PVT settings.


Top-level cell timing derates:
 data_cell derate early:   rise: 0.92    fall: 0.92
 clk_cell derate early:    rise: 0.92    fall: 0.92
 cell_check derate early:   rise: 1.00    fall: 1.00
 data_cell derate  late:   rise: 1.05    fall: 1.05
 clk_cell derate  late:    rise: 1.05    fall: 1.05
 cell_check derate  late:   rise: 1.00    fall: 1.00

Global net timing derates:
 net_data, early:    rise: 0.92    fall: 0.92
 net_data, late:    rise: 1.05    fall: 1.05
 net_clock, early:    rise: 0.92    fall: 0.92
 net_clock, late:    rise: 1.05    fall: 1.05
```

PVT
process 1 · 1.05 V · −40 °C (early / late 동일).

Cell derate
data · clk cell early 0.92 / late 1.05, cell_check 1.00.

Net derate
net_data · net_clock early 0.92 / late 1.05.

Parasitic Parameters

report_parasitic_parameters — corner별 TLU+ 매핑.

report_parasitic_parameters

```
icc2_shell> report_parasitic_parameters

 mapping = /home/project/DB/32nm_samsung/tech/star_rcxt/saed32nm_tf_itf_tluplus.map


Corner: tt0p78v125c
 early_spec = maxTLU
 maxTLU = /home/project/DB/32nm_samsung/tech/star_rcxt/saed32nm_1p9m_Cmax.tluplus
 early_temperature = 125
 late_spec = maxTLU
 maxTLU = /home/project/DB/32nm_samsung/tech/star_rcxt/saed32nm_1p9m_Cmax.tluplus
 late_temperature = 125


Corner: tt1p05vn40c
 early_spec = minTLU
 minTLU = /home/project/DB/32nm_samsung/tech/star_rcxt/saed32nm_1p9m_Cmin.tluplus
 early_temperature = -40
 late_spec = minTLU
 minTLU = /home/project/DB/32nm_samsung/tech/star_rcxt/saed32nm_1p9m_Cmin.tluplus
 late_temperature = -40

library = saed32.nlib
```

tt0p78v125c
early · late maxTLU (Cmax), 125 °C.

tt1p05vn40c
early · late minTLU (Cmin), −40 °C.

02

SECTION

Floorplan & Power Plan

72 × 72 die에 core offset 10, port constraint 보정 후 M3/M4 ring · mesh와 M1 rail 구성.

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floorplan.tcl (1/2)

design rule 재적용 → floorplan → coarse placement → PG → pin 배치.

floorplan.tcl · 1–5

```
#-------------------------------------------------------------------------------
# 1. Open design
#-------------------------------------------------------------------------------
open_lib   nlib/design_setup.nlib
open_block /home/AI23/all_lab_student/tools_TOP/icc2/nlib/design_setup.nlib:TOP.design
link_block

#-------------------------------------------------------------------------------
# 2. Constraints / design rules
#-------------------------------------------------------------------------------
# Same design-rule limits on both scenarios
foreach scn {normal_tt1p05vn40c normal_tt0p78v125c} {
    current_scenario $scn
    set_max_capacitance 10   [all_inputs]
    set_max_capacitance 10   [all_outputs]
    set_max_capacitance 10   [current_design]
    set_max_transition  0.12 [all_inputs]
    set_max_transition  0.12 [all_outputs]
}


#-------------------------------------------------------------------------------
# 3. Floorplan : die {0 0}-{72 72}, core offset 10 on all sides
#-------------------------------------------------------------------------------
initialize_floorplan -boundary {{0 0} {72 72}} \
    -core_offset {10 10 10 10} -control_type die

# Coarse placement to get a first cell distribution inside the floorplan
create_placement -floorplan
report_utilization


#-------------------------------------------------------------------------------
# 4. Legalize and power / ground
#-------------------------------------------------------------------------------
legalize_placement
connect_pg_net
source scripts/pns.tcl            ;# PG mesh / ring / std cell rail

#-------------------------------------------------------------------------------
# 5. Pin placement (top-level ports)
#-------------------------------------------------------------------------------
place_pins -self
```

Design rule

max_capacitance 10 · max_transition 0.12를 all_inputs / all_outputs에 적용.

Floorplan

die {0 0} – {72 72}, core offset 10 (4면).

PG · Pin

legalize → connect_pg_net → pns.tcl → place_pins -self.

floorplan.tcl (2/2)

PG connectivity · DRC · missing via와 utilization 확인.

floorplan.tcl · 6

```
#-------------------------------------------------------------------------------
# 6. Checks
#-------------------------------------------------------------------------------
check_pg_connectivity
check_pg_drc
check_pg_missing_vias
report_utilization

#-------------------------------------------------------------------------------
# 7. Save / export
#-------------------------------------------------------------------------------

check_pg_connectivity > reports/floorplan/report_pg_connectivity_after_floorplan.txt
check_pg_drc > reports/floorplan/report_pg_drc_after_floorplan.txt
check_pg_missing_vias > reports/floorplan/report_pg_missing_vias_after_floorplan.txt
report_utilization > reports/floorplan/report_utilization_after_floorplan.txt

save_lib -as nlib/floorplan.nlib
write_def -include {rows_tracks cells ports specialnets vias} TOP.def
```

Check

check_pg_connectivity · check_pg_drc · check_pg_missing_vias · report_utilization.

Port Constraint — 적용 전 / 후

MY_con의 design rule은 current_design에만 적용 → port에 표시 안 됨, all_inputs / all_outputs에 재적용.

■ 적용 전

Port	MaxTrans	MaxCap	MinCap
VDD	--	--	--
VSS	--	--	--
clk	--	--	--
error[0]	--	--	--
error[1]	--	--	--
error[2]	--	--	--
error[3]	--	--	--
imax1[0]	--	--	--
imax1[1]	--	--	--
imax1[2]	--	--	--
imax1[3]	--	--	--
imax2[0]	--	--	--
imax2[1]	--	--	--
imax2[2]	--	--	--
imax2[3]	--	--	--
imax3[0]	--	--	--
imax3[1]	--	--	--
imax3[2]	--	--	--
imax3[3]	--	--	--
imax4[0]	--	--	--
imax4[1]	--	--	--
imax4[2]	--	--	--
imax4[3]	--	--	--
imin1[0]	--	--	--
imin1[1]	--	--	--
imin1[2]	--	--	--
imin1[3]	--	--	--
imin2[0]	--	--	--
imin2[1]	--	--	--
imin2[2]	--	--	--
imin2[3]	--	--	--
imin3[0]	--	--	--
imin3[1]	--	--	--
imin3[2]	--	--	--
imin3[3]	--	--	--
imin4[0]	--	--	--
imin4[1]	--	--	--
imin4[2]	--	--	--
imin4[3]	--	--	--
max_value[0]	--	--	--
max_value[1]	--	--	--
max_value[2]	--	--	--
max_value[3]	--	--	--
min_value[0]	--	--	--
min_value[1]	--	--	--
min_value[2]	--	--	--
min_value[3]	--	--	--
reset	--	--	--

■ 적용 후 APPLIED

Port	MaxTrans	MaxCap	MinCap
VDD	0.12	10.00	--
VSS	0.12	10.00	--
clk	0.12	10.00	--
error[0]	0.12	10.00	--
error[1]	0.12	10.00	--
error[2]	0.12	10.00	--
error[3]	0.12	10.00	--
imax1[0]	0.12	10.00	--
imax1[1]	0.12	10.00	--
imax1[2]	0.12	10.00	--
imax1[3]	0.12	10.00	--
imax2[0]	0.12	10.00	--
imax2[1]	0.12	10.00	--
imax2[2]	0.12	10.00	--
imax2[3]	0.12	10.00	--
imax3[0]	0.12	10.00	--
imax3[1]	0.12	10.00	--
imax3[2]	0.12	10.00	--
imax3[3]	0.12	10.00	--
imax4[0]	0.12	10.00	--
imax4[1]	0.12	10.00	--
imax4[2]	0.12	10.00	--
imax4[3]	0.12	10.00	--
imin1[0]	0.12	10.00	--
imin1[1]	0.12	10.00	--
imin1[2]	0.12	10.00	--
imin1[3]	0.12	10.00	--
imin2[0]	0.12	10.00	--
imin2[1]	0.12	10.00	--
imin2[2]	0.12	10.00	--
imin2[3]	0.12	10.00	--
imin3[0]	0.12	10.00	--
imin3[1]	0.12	10.00	--
imin3[2]	0.12	10.00	--
imin3[3]	0.12	10.00	--
imin4[0]	0.12	10.00	--
imin4[1]	0.12	10.00	--
imin4[2]	0.12	10.00	--
imin4[3]	0.12	10.00	--
max_value[0]	0.12	10.00	--
max_value[1]	0.12	10.00	--
max_value[2]	0.12	10.00	--
max_value[3]	0.12	10.00	--
min_value[0]	0.12	10.00	--
min_value[1]	0.12	10.00	--
min_value[2]	0.12	10.00	--
min_value[3]	0.12	10.00	--
reset	0.12	10.00	--

0.12

MaxTrans

모든 port

10.00

MaxCap

모든 port

■ **원인**

MY_con의 max_transition 0.12 · max_capacitance 2.0이 [current_design] 기준 → port 별 값 '--'.

■ **적용**

floorplan.tcl에서 all_inputs / all_outputs에 직접 지정 → 전 port에 값 표시.

■ **max_cap 10**

spec 2.0은 실제 load 대비 pessimistic → 10으로 완화.

Floorplan & Coarse Placement

72 × 72 die, core offset 10 — create_placement -floorplan 결과.

layout · coarse placement



72 × 72

Die (μm)

52 × 52

Core (μm)

offset 10 × 4면

Utilization — Floorplan

report_utilization — cell area 44.88 %.

report_utilization · after coarse placement

```
****************************************
Report : report_utilization
Design : TOP
Version: T-2022.03-SP5
Date   : Thu Oct  8 15:11:20 2026
****************************************
Utilization Ratio:                      0.4488
Utilization options:
 - Area calculation based on:           site_row of block TOP
 - Categories of objects excluded:      hard_macros macro_keepouts soft_macros io_cells hard_blockages
Total Area:                             2694.4347
Total Capacity Area:                    2694.4347
Total Area of cells:                    1209.2172
Area of excluded objects:
 - hard_macros         :                0.0000
 - macro_keepouts      :                0.0000
 - soft_macros         :                0.0000
 - io_cells            :                0.0000
 - hard_blockages      :                0.0000

Utilization of site-rows with:
 - Site 'unit':                         0.4488

0.4488
```

0.4488

Utilization

1,209.22

Cell area (μm²)

2,694.43

Total area (μm²)

site row

pns.tcl (1/2)

기존 PG 설정 정리 → ring via master → M3/M4 main mesh.

pns.tcl · 0–1

```
#-------------------------------------------------------------------------------
# 0. Clean up previous PG setup and routes
#-------------------------------------------------------------------------------
remove_pg_strategies -all
remove_pg_patterns -all
remove_pg_regions -all
remove_pg_via_master_rules -all
remove_pg_strategy_via_rules -all
remove_routes -net_types {power ground} -ring -stripe -macro_pin_connect -lib_cell_pin_connect > /dev/null

# Via master used by the ring (M3-M4, 2x2 cut array)
set_pg_via_master_rule VIA_3_to_4 -contact_code VIA34SQ_C -via_array_dimension {2 2}

#-------------------------------------------------------------------------------
# 1. Main power mesh (M3 horizontal / M4 vertical)
#    width 0.4, pitch 13.376, VSS/VDD interleaved, trimmed at the boundary
#-------------------------------------------------------------------------------
create_pg_mesh_pattern P_m3m4 \
        -layers { \
                { {horizontal_layer: M3} {width: 0.4} {spacing: interleaving} {pitch: 13.376} {offset: 0.75} {trim : true} } \
                { {vertical_layer: M4}   {width: 0.4} {spacing: interleaving} {pitch: 13.376} {offset: 0.75}  {trim : true} } \
                } \
        -via_rule { {intersection: adjacent} {via_master : {default}} }

# Apply the mesh over the core area, start at (10,10), extend to the design boundary and create pins
set_pg_strategy S_vddvss \
        -core \
        -pattern   { {name: P_m3m4} {nets:{VSS VDD}} {offset_start: {10 10}} } \
        -extension { {{stop:design_boundary_and_generate_pin}} }

compile_pg -strategies {S_vddvss}
```

Ring via
VIA_3_to_4 = VIA34SQ_C, array {2 2} — spec ring via와 동일.

Mesh
M3 H / M4 V, width 0.4 · pitch 13.376 · offset 0.75, VSS / VDD interleave.

Extension
core (10, 10)부터 design boundary까지 확장, boundary에 PG pin 생성.

pns.tcl (2/2)

core ring → M1 std cell rail → M1 – M4 stacked via → PG check.

pns.tcl · 2–4

```
#-------------------------------------------------------------------------------
# 2. Power ring around the core (M3 horizontal / M4 vertical, width 1)
#-------------------------------------------------------------------------------
suppress_message PGR-599

create_pg_ring_pattern P_m3m4_ring \
        -horizontal_layer M3 -horizontal_width {1} \
        -vertical_layer   M4 -vertical_width   {1} \
        -horizontal_spacing {0.3} -vertical_spacing {0.3} \
        -corner_bridge false \
        -via_rule {{via_master : {VIA_3_to_4}}}

set_pg_strategy S_m3m4_ring -core \
        -pattern { {pattern: P_m3m4_ring} {nets: {VSS VDD}} {offset: 1 1}}

# Connect the ring to the existing mesh straps (M3 and M4 crossings)
set_pg_strategy_via_rule S_ring_vias -via_rule { \
        {{strategies: {S_m3m4_ring}} {layers: {M3}}} {existing: {strap }}{via_master: {VIA_3_to_4}}} \
        {{strategies: {S_m3m4_ring}} {layers: {M4}}} {existing: {strap }}{via_master: {VIA_3_to_4}}} \
}

compile_pg -strategies {S_m3m4_ring} -via_rule S_ring_vias

#-------------------------------------------------------------------------------
# 3. Standard cell rail (M1) and M1 -> M4 stacked via
#-------------------------------------------------------------------------------
# Per-layer via masters (3x1 cut arrays) stacked to reach M4 from M1
set_pg_via_master_rule pgvia12 -contact_code VIA12SQ -via_array_dimension {3 1}
set_pg_via_master_rule pgvia23 -contact_code VIA23SQ -via_array_dimension {3 1}
set_pg_via_master_rule pgvia34 -contact_code VIA34SQ -via_array_dimension {3 1}

create_pg_std_cell_conn_pattern P_std_cell_rail

# Rails run along every row, extended up to the outermost ring on all four sides
set_pg_strategy S_std_cell_rail_VSS_VDD \
        -core \
        -pattern {{pattern: P_std_cell_rail}{nets: {VSS VDD}}} \
        -extension {{stop: outermost_ring}{direction: L B R T }}

# Stack VIA12/23/34 where the M1 rail crosses an existing M4 strap; no vias elsewhere
set_pg_strategy_via_rule S_via_stdcellrail -via_rule { \
    { {{strategies: {S_std_cell_rail_VSS_VDD}}{layers: {M1}}} \
      {{existing: {strap}}{layers: {M4}}} \
      {via_master: {pgvia12 pgvia23 pgvia34}} } \
    { {intersection: undefined}{via_master: NIL} } \
}

compile_pg -strategies {S_std_cell_rail_VSS_VDD} -via_rule S_via_stdcellrail
```

Ring
M3 / M4 width 1 · spacing 0.3, mesh strap 교차점에 VIA_3_to_4.

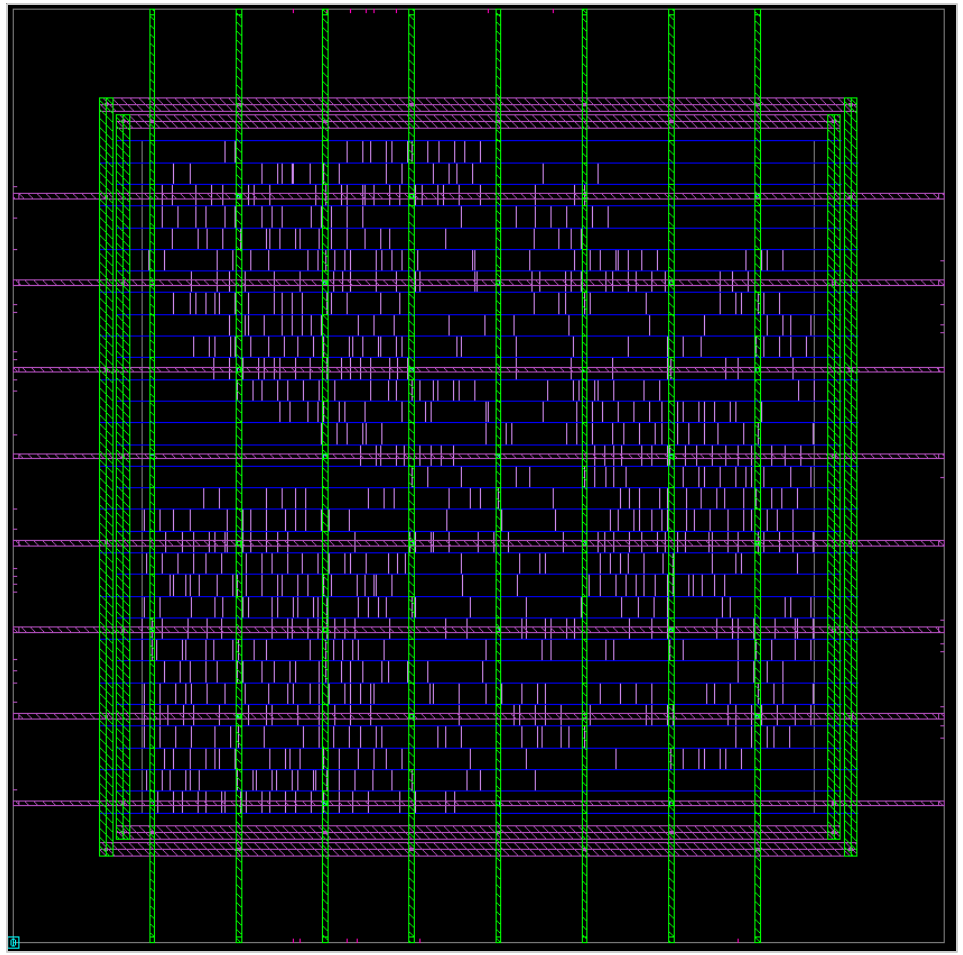
Rail
M1 std cell rail을 4방향 outermost ring까지 확장.

Stacked via
rail × M4 strap 교차점에만 VIA12 · 23 · 34 (3 × 1).

Ring & Mesh

core 둘레 M3/M4 ring, core 위 M3 · M4 mesh, row마다 M1 rail.

layout · after PNS



Ring
core를 감싸는 M3 (H) · M4 (V) ring, width 1.

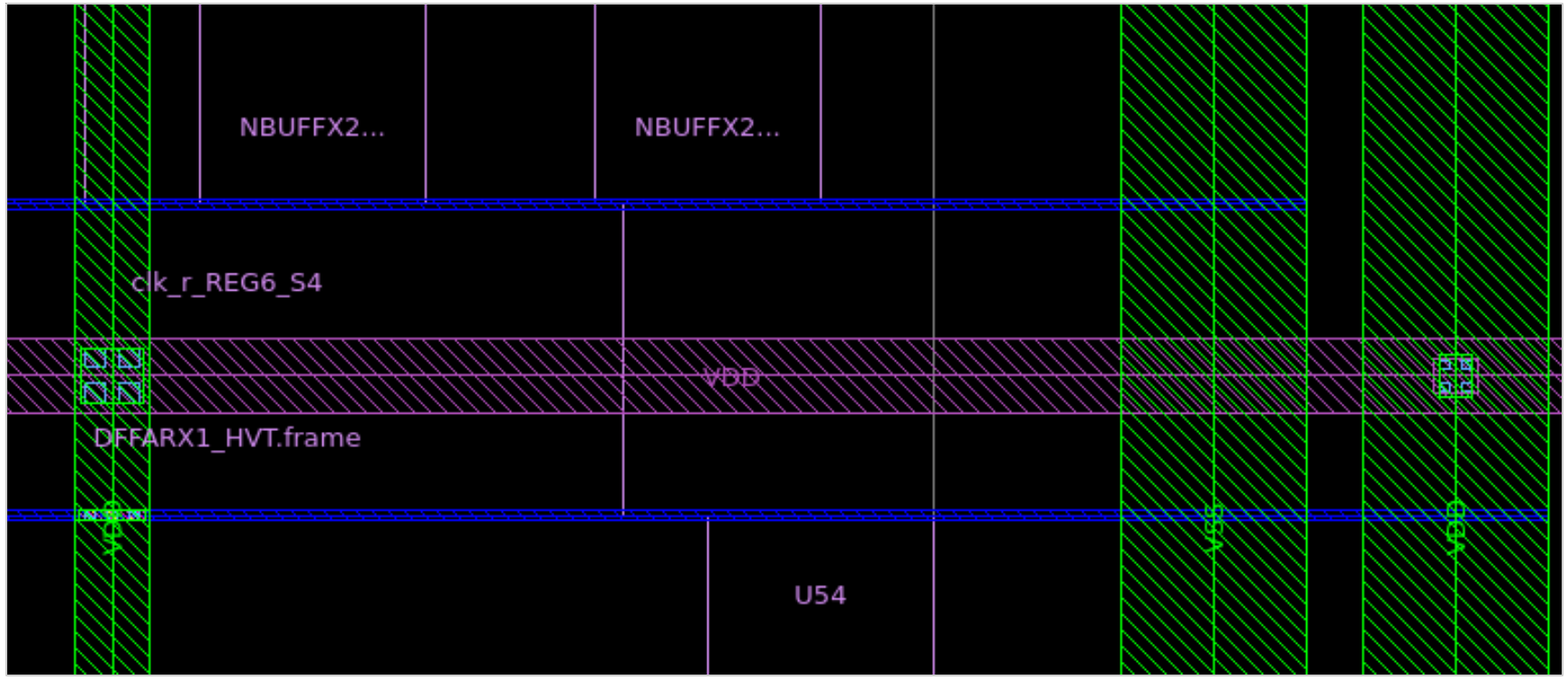
Mesh
M3 · M4 strap, boundary까지 확장 — pitch 13.376.

Rail
std cell row를 따라 M1 VDD / VSS rail.

PG Via

rail – strap stacked via와 ring · mesh의 M3 – M4 via.

layout · via after PNS



Rail – Strap
M1 rail × M4 strap 교차점에 VIA12 / 23 / 34 stacked via.

Ring
top ring의 M3 – M4를 VIA_3_to_4 (VIA34SQ_C 2 × 2)로 연결.

Mesh
mesh M3 strap과 M4 strap 교차점에 M3 – M4 via.

PG Connectivity — Floorplan

check_pg_connectivity — VDD / VSS 모두 floating 0.

check_pg_connectivity · after PNS

PASS

```
icc2_shell> check_pg_connectivity
START_CMD: check_pg_connectivity CPU:     95 s ( 0.03 hr) ELAPSE:    999 s ( 0.28 hr) MEM-PEAK
Checking secondary net through power switch is enabled.
Secondary net will be checked together from primary net. They will be treated as the same net
Primary Net : VDD    Secondary Net:
Primary Net : VSS    Secondary Net:
Loading cell instances...
Number of Standard Cells: 448
Number of Macro Cells: 0
Number of IO Pad Cells: 0
Number of Blocks: 0
Loading P/G wires and vias...
Number of VDD Wires: 28
Number of VDD Vias: 228
Number of VDD Terminals: 16
**************Verify net VDD connectivity*****************
  Number of floating wires: 0
  Number of floating vias: 0
  Number of floating std cells: 0
  Number of floating hard macros: 0
  Number of floating I/O pads: 0
  Number of floating terminals: 0
  Number of floating hierarchical blocks: 0
************************************************************
Loading cell instances...
Loading P/G wires and vias...
Number of VSS Wires: 28
Number of VSS Vias: 228
Number of VSS Terminals: 16
**************Verify net VSS connectivity*****************
  Number of floating wires: 0
  Number of floating vias: 0
  Number of floating std cells: 0
  Number of floating hard macros: 0
  Number of floating I/O pads: 0
  Number of floating terminals: 0
  Number of floating hierarchical blocks: 0
************************************************************
```

0

Floating

wire · via · cell · terminal

28 / 228

Wires / Vias

VDD · VSS 각각

PG DRC & Missing Via

check_pg_drc · check_pg_missing_vias — 위반 없음.

check_pg_drc · check_pg_missing_vias **PASS**

```
icc2_shell> check_pg_drc
START_CMD: check_pg_drc          CPU:    100 s ( 0
Command check_pg_drc started  at Thu Oct  8 15:2
Command check_pg_drc finished at Thu Oct  8 15:2
CPU usage for check_pg_drc: 1.80 seconds ( 0.00
Elapsed time for check_pg_drc: 0.25 seconds ( 0.
No errors found.
END_CMD: check_pg_drc            CPU:    102 s ( 0
icc2_shell> check_pg_missing_vias
START_CMD: check_pg_missing_vias CPU:    102 s (
Check net VDD vias...
Number of missing vias: 0
Checking net VDD vias took 0 seconds.
Check net VSS vias...
Number of missing vias: 0
Checking net VSS vias took 0 seconds.
Overall runtime: 0 seconds.
END_CMD: check_pg_missing_vias CPU:    102 s ( 0
```

No errors

check_pg_drc

0 / 0

Missing vias

VDD / VSS

03

SECTION

Place

pre-placement check와 congestion 예측 후 place_opt → refine_opt로 배치 최적화.

place.tcl (1/2)

pre-placement check → Vth · fanout 제약 → congestion 예측 → CCD off.

place.tcl · 0–3

```
#-------------------------------------------------------------------------------
# 0. Open the floorplan design and check placement readiness
#-------------------------------------------------------------------------------
open_lib   nlib/floorplan.nlib/
open_block /home/AI23/all_lab_student/tools_TOP/icc2/nlib/floorplan.nlib:TOP.design
link_block

check_design -checks pre_placement_stage

#-------------------------------------------------------------------------------
# 1. Optimization constraints
#-------------------------------------------------------------------------------
# Multi-Vth : LVT cells limited to 20% of the design (counted by cell number)
set_threshold_voltage_group_type -type low_vt saed32cell_lvt
set_multi_vth_constraint -low_vt_percentage 20 -cost cell_count

# Max fanout used by optimization
set_app_options -list {opt.common.max_fanout 12}

#-------------------------------------------------------------------------------
# 2. Congestion estimate before placement (global route, map only)
#-------------------------------------------------------------------------------
route_global -congestion_map_only true -effort_level low
report_congestion

#-------------------------------------------------------------------------------
# 3. Turn CCD / local skew optimization off
#    (balanced tree, no useful skew; the cts.* options are kept in the block
#     and take effect later in clock_opt)
#-------------------------------------------------------------------------------
set_app_options -list {clock_opt.flow.enable_ccd {false}}
set_app_options -list {place_opt.flow.enable_ccd {false}}
set_app_options -list {cts.optimize.enable_area_recovery_in_local_skew {false}}
set_app_options -list {cts.common.enable_auto_skew_target_for_local_skew {false}}
```

Multi-Vth
LVT cell_count 20 % 제약 — spec LVT < 20 %.

Congestion
route_global -congestion_map_only로 배치 전 예측.

CCD off
balanced tree 목적 — CCD · local skew 옵션 off.

place.tcl (2/2)

place_opt → refine_opt, 단계마다 congestion · QoR · constraint 확인.

place.tcl · 4–5

```
#--------------------------------------------------------------------------
# 4. Placement optimization (initial_place -> initial_drc -> initial_opto
#                            -> final_place -> final_opto)
#--------------------------------------------------------------------------
place_opt
report_congestion
report_qor -summary -significant_digits 5
report_qor
report_constraints -all_violators -scenarios {normal_tt0p78v125c normal_tt1p05vn40c}
report_utilization

#--------------------------------------------------------------------------
# 5. refine_opt
#--------------------------------------------------------------------------
refine_opt
report_congestion
report_qor -summary
report_constraints -all_violators -scenarios {normal_tt0p78v125c normal_tt1p05vn40c}
report_utilization

#--------------------------------------------------------------------------
# 6. Save
#--------------------------------------------------------------------------
report_congestion > reports/place/report_congestion_after_place.txt
report_qor -summary -significant_digits 5 > reports/place/report_qor_summary_after_place.txt
report_qor > reports/place/report_qor_after_place.txt
report_constraints -all_violators -scenarios {normal_tt0p78v125c normal_tt1p05vn40c} > reports/place/report_constraints_after_place.txt
report_utilization > reports/place/report_utilization_after_place.txt
save_lib -as nlib/place.nlib
```

place_opt
initial_place → initial_drc → initial_opto → final_place → final_opto.

refine_opt
path 기반 최적화 — 현재는 사라졌지만 legacy로 남아 사용 가능해 이를 사용하여 최적화.

Pre-placement Check

check_design pre_placement_stage — error 0, warning은 모두 원인 확인.

check_design · pre_placement_stage

0 ERRORS

```
****************************************
 Report : check_design
 Options: { pre_placement_stage }
 Design : TOP
 Version: T-2022.03-SP5
 Date   : Thu Oct  8 15:26:58 2026
****************************************

Running mega-check 'pre_placement_stage':
    Running atomic-check 'design_mismatch'
    Running atomic-check 'scan_chain'
    Running atomic-check 'mv_design'
    Running atomic-check 'rp_constraints'
    Running atomic-check 'timing'
    Running atomic-check 'hier_pre_placement'

  *** EMS Message summary ***
  ------------------------------------------------------------------------------------------------------
  Rule         Type   Count      Message
  ------------------------------------------------------------------------------------------------------
  DFT-011      Info   1          The design has no scan chain defined in the scandef.
  TCK-001      Warn   60         The reported endpoint '%endpoint' is unconstrained. Reason: '%re...
  TCK-012      Warn   34         The input port '%port' has no clock_relative delay specified. Mo...
  ------------------------------------------------------------------------------------------------------
  Total 95 EMS messages : 0 errors, 94 warnings, 1 info.
  ------------------------------------------------------------------------------------------------------

  *** Non-EMS message summary ***
  ------------------------------------------------------------------------------------------------------
  Rule         Type   Count      Message
  ------------------------------------------------------------------------------------------------------
  NDMUI-173    Info   1          There are no relative placement groups in the design.
  NEX-001      Warn   6          Technology layer '%s' setting '%s' is not valid
  PVT-032      Info   2          Corner %s: no PVT mismatches.
  ------------------------------------------------------------------------------------------------------
  Total 9 non-EMS messages : 0 errors, 6 warnings, 3 info.
  ------------------------------------------------------------------------------------------------------
```

TCK-001 (60)
async set/reset pin(SETB · RSTB) endpoint, reset input delay 미지정으로 no arrival path → unconstrained (의도됨, register 60개와 일치).

TCK-012 (34)
input port에 clock 기준 input delay 없음 → tt1p05vn40c I/O constraint 미적용. reset은 의도적 미지정.

NEX-001 (6)
technology layer 설정값이 유효하지 않음 (man: tech setup 확인 권고). 미사용 layer 관련.

TCK-001 / TCK-012

TCK-001은 async set/reset endpoint(의도됨), TCK-012는 tt1p05vn40c input delay 미적용 — No paths로 확인.

TCK-012 (34) · reset + imax* · imin* 32 port

TCK-012	1	active	The input port 'reset' has no clock_relative delay specified. Mode:'normal', Corn...
TCK-012	2	active	The input port 'imax1[3]' has no clock_relative delay specified. Mode:'normal', ...
TCK-012	3	active	The input port 'imax1[2]' has no clock_relative delay specified. Mode:'normal', ...
TCK-012	4	active	The input port 'imax1[1]' has no clock_relative delay specified. Mode:'normal', ...
TCK-012	5	active	The input port 'imax1[0]' has no clock_relative delay specified. Mode:'normal', ...
TCK-012	6	active	The input port 'imax2[3]' has no clock_relative delay specified. Mode:'normal', ...
TCK-012	7	active	The input port 'imax2[2]' has no clock_relative delay specified. Mode:'normal', ...
TCK-012	8	active	The input port 'imax2[1]' has no clock_relative delay specified. Mode:'normal', ...
TCK-012	9	active	The input port 'imax2[0]' has no clock_relative delay specified. Mode:'normal', ...
TCK-012	10	active	The input port 'imax3[3]' has no clock_relative delay specified. Mode:'normal', ...
TCK-012	11	active	The input port 'imax3[2]' has no clock_relative delay specified. Mode:'normal', ...
TCK-012	12	active	The input port 'imax3[1]' has no clock_relative delay specified. Mode:'normal', ...
TCK-012	13	active	The input port 'imax3[0]' has no clock_relative delay specified. Mode:'normal', ...
TCK-012	14	active	The input port 'imax4[3]' has no clock_relative delay specified. Mode:'normal', ...
TCK-012	15	active	The input port 'imax4[2]' has no clock_relative delay specified. Mode:'normal', ...
TCK-012	16	active	The input port 'imax4[1]' has no clock_relative delay specified. Mode:'normal', ...
TCK-012	17	active	The input port 'imax4[0]' has no clock_relative delay specified. Mode:'normal', ...
TCK-012	18	active	The input port 'imin1[3]' has no clock_relative delay specified. Mode:'normal', C...
TCK-012	19	active	The input port 'imin1[2]' has no clock_relative delay specified. Mode:'normal', C...
TCK-012	20	active	The input port 'imin1[1]' has no clock_relative delay specified. Mode:'normal', C...
TCK-012	21	active	The input port 'imin1[0]' has no clock_relative delay specified. Mode:'normal', C...
TCK-012	22	active	The input port 'imin2[3]' has no clock_relative delay specified. Mode:'normal', C...
TCK-012	23	active	The input port 'imin2[2]' has no clock_relative delay specified. Mode:'normal', C...
TCK-012	24	active	The input port 'imin2[1]' has no clock_relative delay specified. Mode:'normal', C...
TCK-012	25	active	The input port 'imin2[0]' has no clock_relative delay specified. Mode:'normal', C...
TCK-012	26	active	The input port 'imin3[3]' has no clock_relative delay specified. Mode:'normal', C...
TCK-012	27	active	The input port 'imin3[2]' has no clock_relative delay specified. Mode:'normal', C...
TCK-012	28	active	The input port 'imin3[1]' has no clock_relative delay specified. Mode:'normal', C...
TCK-012	29	active	The input port 'imin3[0]' has no clock_relative delay specified. Mode:'normal', C...
TCK-012	30	active	The input port 'imin4[3]' has no clock_relative delay specified. Mode:'normal', C...
TCK-012	31	active	The input port 'imin4[2]' has no clock_relative delay specified. Mode:'normal', C...
TCK-012	32	active	The input port 'imin4[1]' has no clock_relative delay specified. Mode:'normal', C...
TCK-012	33	active	The input port 'imin4[0]' has no clock_relative delay specified. Mode:'normal', C...
TCK-012	94	active	The input port 'reset' has no clock_relative delay specified. Mode:'normal', Corn...

report_timing -delay_type min

NO PATHS

```
****************************************
Report : timing
        -path_type full
        -delay_type min
        -max_paths 1
        -report_by design
Design : TOP
Version: T-2022.03-SP5
Date   : Mon Oct 12 10:35:22 2026
****************************************
Warning: From_pin 'reset' is not a timing startpoint and will be ignored.
No paths.
END_CMD: report_timing          CPU:     30 s ( 0.01 hr) ELAPSE:     319 s (
1
```

TCK-001 (60) · SETB · RSTB endpoint

INTENDED

TCK-001	34	active	The reported endpoint 'imax_1/clk_r_REG18_S1/RSTB' is unconstrained. Reason:...
TCK-001	35	active	The reported endpoint 'imax_1/clk_r_REG19_S1/RSTB' is unconstrained. Reason:...
TCK-001	36	active	The reported endpoint 'imax_1/clk_r_REG20_S1/RSTB' is unconstrained. Reason:...
TCK-001	37	active	The reported endpoint 'imax_1/clk_r_REG22_S1_IP/SETB' is unconstrained. Reas...
TCK-001	38	active	The reported endpoint 'imax_1/clk_r_REG23_S1_IP/SETB' is unconstrained. Reas...
TCK-001	39	active	The reported endpoint 'imax_1/clk_r_REG24_S1_IP/SETB' is unconstrained. Reas...
TCK-001	40	active	The reported endpoint 'imax_1/clk_r_REG26_S1_IP/SETB' is unconstrained. Reas...
TCK-001	41	active	The reported endpoint 'imax_1/clk_r_REG27_S1_IP/SETB' is unconstrained. Reas...
TCK-001	42	active	The reported endpoint 'imax_1/clk_r_REG28_S1/RSTB' is unconstrained. Reason:...
TCK-001	43	active	The reported endpoint 'imax_1/clk_r_REG29_S1_IP/SETB' is unconstrained. Reas...
TCK-001	44	active	The reported endpoint 'imax_1/clk_r_REG30_S1_IP/SETB' is unconstrained. Reas...
TCK-001	45	active	The reported endpoint 'imax_1/clk_r_REG31_S1_IP/SETB' is unconstrained. Reas...
TCK-001	46	active	The reported endpoint 'imax_1/clk_r_REG7_S1/SETB' is unconstrained. Reason: '...
TCK-001	47	active	The reported endpoint 'imax_1/clk_r_REG21_S1/SETB' is unconstrained. Reason:...
TCK-001	48	active	The reported endpoint 'imax_1/clk_r_REG25_S1/SETB' is unconstrained. Reason:...
TCK-001	49	active	The reported endpoint 'imin_1/clk_r_REG32_S1_IP/SETB' is unconstrained. Reaso...
TCK-001	50	active	The reported endpoint 'imin_1/clk_r_REG51_S1_IP/SETB' is unconstrained. Reaso...
TCK-001	51	active	The reported endpoint 'imin_1/clk_r_REG54_S1/RSTB' is unconstrained. Reason: ...

Pre-placement Check — 재확인

tt1p05vn40c I/O constraint 재적용 후 — TCK-012 34 → 2, NEX-001 사라짐.

check_design · pre_placement_stage · 재확인

0 ERRORS

```
****************************************
 Report : check_design
 Options: { pre_placement_stage }
 Design : TOP
 Version: T-2022.03-SP5
 Date   : Mon Oct 12 10:36:43 2026
****************************************

Running mega-check 'pre_placement_stage':
    Running atomic-check 'design_mismatch'
    Running atomic-check 'scan_chain'
    Running atomic-check 'mv_design'
    Running atomic-check 'rp_constraints'
    Running atomic-check 'timing'
    Running atomic-check 'hier_pre_placement'

  *** EMS Message summary ***
  ------------------------------------------------------------------------------------------------
  Rule         Type    Count      Message
  ------------------------------------------------------------------------------------------------
  DFT-011      Info    1          The design has no scan chain defined in the scandef.
  TCK-001      Warn    60         The reported endpoint '%endpoint' is unconstrained. Reason: '%re...
  TCK-012      Warn    2          The input port '%port' has no clock_relative delay specified. Mo...
  ------------------------------------------------------------------------------------------------
  Total 63 EMS messages : 0 errors, 62 warnings, 1 info.
  ------------------------------------------------------------------------------------------------

  *** Non-EMS message summary ***
  ------------------------------------------------------------------------------------------------
  Rule         Type    Count      Message
  ------------------------------------------------------------------------------------------------
  NDMUI-173    Info    1          There are no relative placement groups in the design.
  ------------------------------------------------------------------------------------------------
  Total 1 non-EMS messages : 0 errors, 0 warnings, 1 info.
  ------------------------------------------------------------------------------------------------
```

34 → 2

TCK-012

reset × 2 scenario · 의도됨

60

TCK-001

변화 없음 · 의도됨

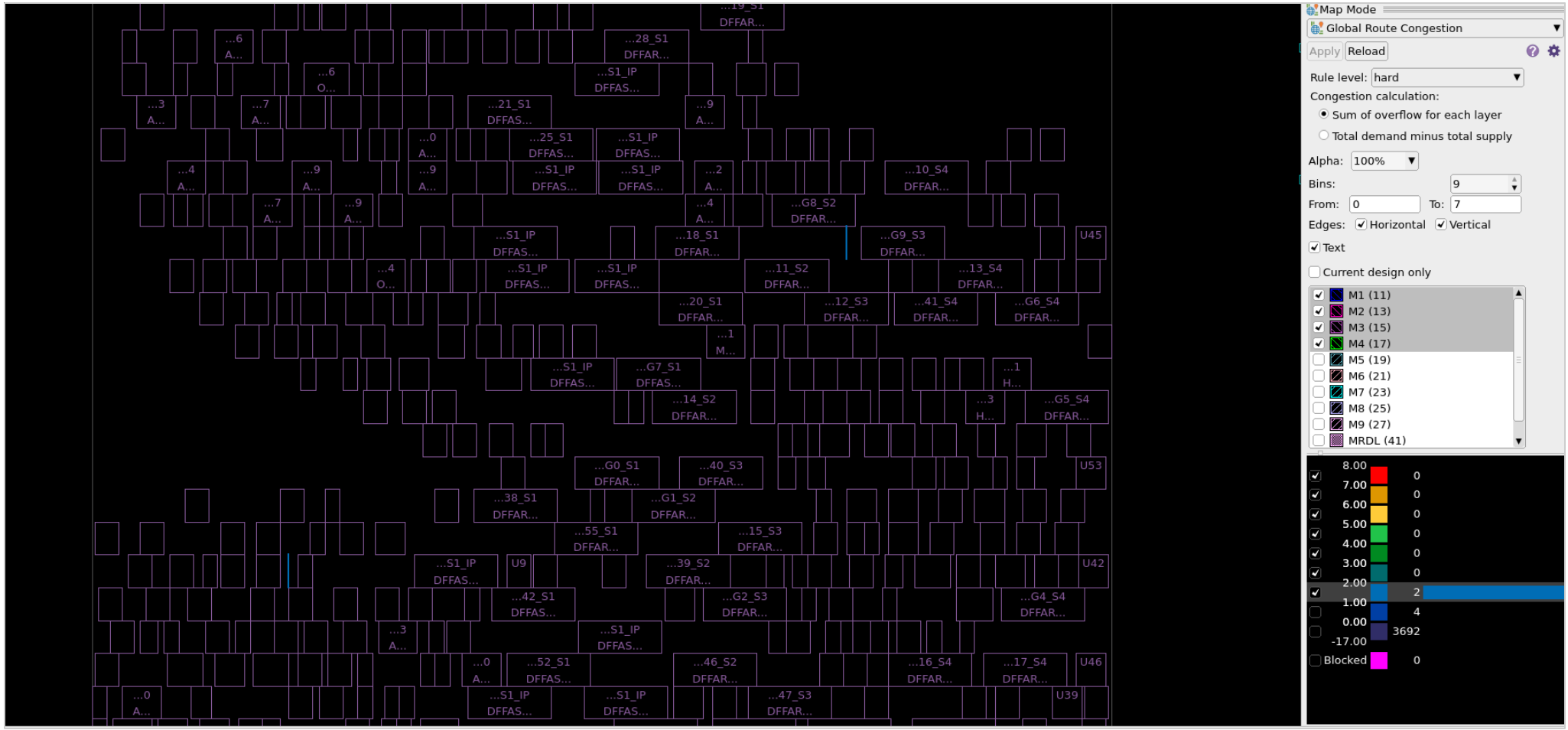
조치

scenario 전환 후 MY_CON.con 재source → imax* · imin* input delay 적용.

Congestion Map — Before Place

global route congestion map (M1 – M4) — overflow 영역 거의 없음.

Global Route Congestion · map only



Congestion Report — Before Place

report_congestion — overflow GRC 2개 (0.05 %).

report_congestion · before place

```
****************************************
Report : congestion
Design : TOP
Version: T-2022.03-SP5
Date   : Thu Oct  8 15:27:58 2026
****************************************

Layer      |     overflow      |                  # GRCs has
Name       |  total  |   max   | overflow (%)       | max overflow
-------------------------------------------------------------------
Both Dirs  |        2 |      1 |        2  ( 0.05%) |        2
H routing  |        2 |      1 |        2  ( 0.11%) |        2
V routing  |        0 |      0 |        0  ( 0.00%) |        0
```

2
Overflow total
max 1

0.05 %
GRC overflow
Both Dirs

0.11 % / 0 %
H / V routing

place_opt — QoR & Congestion

place_opt 직후 congestion 0, setup 만족 · 두 scenario hold 각 4 path 위반.

report_congestion · report_qor -summary

```
****************************************
Report : congestion
Design : TOP
Version: T-2022.03-SP5
Date   : Mon Oct 12 11:03:39 2026
****************************************

Layer     |    overflow     |             # GRCs has
Name      |  total  |  max  | overflow (%)      | max overflow
------------------------------------------------------------
Both Dirs |       0 |     0 |       0  ( 0.00%) |       0
H routing |       0 |     0 |       0  ( 0.00%) |       0
V routing |       0 |     0 |       0  ( 0.00%) |       0

END_CMD: report_congestion     CPU:     94 s ( 0.03 hr) ELAPSE:    170 s ( 0
1
icc2_shell> report_qor -summary -significant_digits 5
****************************************
Report : qor
        -summary
Design : TOP
Version: T-2022.03-SP5
Date   : Mon Oct 12 11:04:37 2026
****************************************

Timing
------------------------------------------------------------------------------
Context                                 WNS            TNS            NVE
------------------------------------------------------------------------------
normal_tt0p78v125c (Setup)          0.00092        0.00000              0
normal_tt1p05vn40c (Setup)          0.20430        0.00000              0
Design             (Setup)          0.00092        0.00000              0

normal_tt0p78v125c (Hold)          -0.04709       -0.14932              4
normal_tt1p05vn40c (Hold)          -0.03482       -0.11876              4
Design             (Hold)          -0.04709       -0.14932              4
------------------------------------------------------------------------------

Miscellaneous
------------------------------------------------------------------------------
Cell Area (netlist):                         863.58131
Cell Area (netlist and physical only):       863.58100
Nets with DRC Violations:        0
```

+0.00092

Setup WNS

NVE 0

−0.04709

Hold WNS

tt0p78v125c · 4 path

0

Overflow

863.58

Cell area (μm²)

place_opt — Constraints

hold endpoint 8개 (imax_1 REG18 · 19 · 20 · 28 × 2 scenario), design rule 위반 0.

report_constraints -all_violators · after place_opt

```
Endpoint                        Path Delay     Path Required       CRP    Slack Group    Scenario
-------------------------------------------------------------------------------------------------------
imax_1/clk_r_REG28_S1/D (DFFARX2_HVT)
                                  0.30 f         0.35           0.00    -0.05   **in2reg_default**
                                                                                 normal_tt0p78v125c
imax_1/clk_r_REG20_S1/D (DFFARX2_HVT)
                                  0.30 f         0.34           0.00    -0.04   **in2reg_default**
                                                                                 normal_tt0p78v125c
imax_1/clk_r_REG19_S1/D (DFFARX2_HVT)
                                  0.31 f         0.34           0.00    -0.04   **in2reg_default**
                                                                                 normal_tt0p78v125c
imax_1/clk_r_REG18_S1/D (DFFARX2_HVT)
                                  0.31 f         0.34           0.00    -0.03   **in2reg_default**
                                                                                 normal_tt0p78v125c
imax_1/clk_r_REG28_S1/D (DFFARX2_HVT)
                                  0.29 f         0.33           0.00    -0.03   **in2reg_default**
                                                                                 normal_tt1p05vn40c
imax_1/clk_r_REG20_S1/D (DFFARX2_HVT)
                                  0.30 f         0.33           0.00    -0.03   **in2reg_default**
                                                                                 normal_tt1p05vn40c
imax_1/clk_r_REG19_S1/D (DFFARX2_HVT)
                                  0.30 f         0.33           0.00    -0.03   **in2reg_default**
                                                                                 normal_tt1p05vn40c
imax_1/clk_r_REG18_S1/D (DFFARX2_HVT)
                                  0.30 f         0.32           0.00    -0.02   **in2reg_default**
                                                                                 normal_tt1p05vn40c


   Mode: normal Corner: tt0p78v125c
   Scenario: normal_tt0p78v125c
  ---------------------------------------------------------------------------
   Number of max_transition violation(s): 0

   Mode: normal Corner: tt1p05vn40c
   Scenario: normal_tt1p05vn40c
  ---------------------------------------------------------------------------
   Number of max_transition violation(s): 0

   Mode: normal Corner: tt0p78v125c
   Scenario: normal_tt0p78v125c
  ---------------------------------------------------------------------------
   Number of max_capacitance violation(s): 0

   Mode: normal Corner: tt1p05vn40c
   Scenario: normal_tt1p05vn40c
  ---------------------------------------------------------------------------
   Number of max_capacitance violation(s): 0


   Mode: normal Corner: tt0p78v125c
   Scenario: normal_tt0p78v125c
  ---------------------------------------------------------------------------
   Number of min_capacitance violation(s): 0

   Mode: normal Corner: tt1p05vn40c
   Scenario: normal_tt1p05vn40c
  ---------------------------------------------------------------------------
   Number of min_capacitance violation(s): 0
```

8

Hold endpoints

−0.05 ~ −0.02

0

DRC

trans · cap

place_opt — Vth Ratio

report_threshold_voltage_group — LVT 3 / 274 cell (1.09 %).

report_threshold_voltage_group · after place_opt

```
****************************************
Report : threshold_voltage_group
Design : TOP
Version: T-2022.03-SP5
Date   : Mon Oct 12 11:06:22 2026
****************************************

Cell Count Report
-----------------------------------------------------------------------------------------------------------------------------------------------------------------

Group Names                     All (%)       Repeaters (%)   Combinational (%)      Registers (%)     Sequentials (%)   Clock_network (%)          Macro (%)   Physical_only (%)
-----------------------------------------------------------------------------------------------------------------------------------------------------------------
saed32cell_hvt            115  (41.97%)       71  (25.91%)       20   (7.30%)       24   (8.76%)        0   (0.00%)        0   (0.00%)        0   (0.00%)        0   (0.00%)
saed32cell_lvt              3   (1.09%)        2   (0.73%)        1   (0.36%)        0   (0.00%)        0   (0.00%)        0   (0.00%)        0   (0.00%)        0   (0.00%)  L
saed32cell_svt            156  (56.93%)       32  (11.68%)       88  (32.12%)       36  (13.14%)        0   (0.00%)        0   (0.00%)        0   (0.00%)        0   (0.00%)
-----------------------------------------------------------------------------------------------------------------------------------------------------------------
Total                     274 (100.00%)      105  (38.32%)      109  (39.78%)       60  (21.90%)        0   (0.00%)        0   (0.00%)        0   (0.00%)        0   (0.00%)
Low vth groups              3   (1.09%)        2   (0.73%)        1   (0.36%)        0   (0.00%)        0   (0.00%)        0   (0.00%)        0   (0.00%)        0   (0.00%)
-----------------------------------------------------------------------------------------------------------------------------------------------------------------


Cell Area Report
-----------------------------------------------------------------------------------------------------------------------------------------------------------------

Group Names                     All (%)       Repeaters (%)   Combinational (%)      Registers (%)     Sequentials (%)   Clock_network (%)          Macro (%)   Physical_only (%)
-----------------------------------------------------------------------------------------------------------------------------------------------------------------
saed32cell_hvt         331.66  (38.40%)    106.23  (12.30%)     44.73   (5.18%)    180.70  (20.92%)     0.00   (0.00%)     0.00   (0.00%)     0.00   (0.00%)     0.00   (0.00%)
saed32cell_lvt           4.57   (0.53%)      2.54   (0.29%)      2.03   (0.24%)      0.00   (0.00%)     0.00   (0.00%)     0.00   (0.00%)     0.00   (0.00%)     0.00   (0.00%)  L
saed32cell_svt         527.35  (61.07%)     48.03   (5.56%)    223.14  (25.84%)    256.18  (29.66%)     0.00   (0.00%)     0.00   (0.00%)     0.00   (0.00%)     0.00   (0.00%)
-----------------------------------------------------------------------------------------------------------------------------------------------------------------
Total                  863.58 (100.00%)    156.81  (18.16%)    269.90  (31.25%)    436.87  (50.59%)     0.00   (0.00%)     0.00   (0.00%)     0.00   (0.00%)     0.00   (0.00%)
Low vth groups           4.57   (0.53%)      2.54   (0.29%)      2.03   (0.24%)      0.00   (0.00%)     0.00   (0.00%)     0.00   (0.00%)     0.00   (0.00%)     0.00   (0.00%)
-----------------------------------------------------------------------------------------------------------------------------------------------------------------
```

1.09 %
LVT
3 / 274

41.97 %
HVT
115

56.93 %
SVT
156

refine_opt — QoR & Congestion

refine_opt 후 hold 대부분 해소 — tt1p05vn40c 1 path 잔여, cell area 863.58 → 829.02.

report_congestion · report_qor -summary

```
****************************************
Report : congestion
Design : TOP
Version: T-2022.03-SP5
Date   : Mon Oct 12 11:07:04 2026
****************************************

Layer     |    overflow     |             # GRCs has
Name      |  total  |  max  | overflow (%)      | max overflow
------------------------------------------------------------
Both Dirs |       1 |     1 |       1  ( 0.03%) |       1
H routing |       1 |     1 |       1  ( 0.05%) |       1
V routing |       0 |     0 |       0  ( 0.00%) |       0

END_CMD: report_congestion     CPU:    117 s ( 0.03 hr) ELAPSE:    376 s ( 0
1
icc2_shell> report_qor -summary -significant_digits 5
****************************************
Report : qor
        -summary
Design : TOP
Version: T-2022.03-SP5
Date   : Mon Oct 12 11:07:10 2026
****************************************

Timing
------------------------------------------------------------------------------
Context                                 WNS            TNS            NVE
------------------------------------------------------------------------------
normal_tt0p78v125c (Setup)          0.01828        0.00000              0
normal_tt1p05vn40c (Setup)          0.20474        0.00000              0
Design             (Setup)          0.01828        0.00000              0

normal_tt0p78v125c (Hold)           0.03001        0.00000              0
normal_tt1p05vn40c (Hold)          -0.00037       -0.00037              1
Design             (Hold)          -0.00037       -0.00037              1
------------------------------------------------------------------------------

Miscellaneous
------------------------------------------------------------------------------
Cell Area (netlist):                         829.01773
Cell Area (netlist and physical only):       829.01800
Nets with DRC Violations:        0
```

+0.01828

Setup WNS

NVE 0

−0.00037

Hold WNS

tt1p05vn40c · 1 path

1 (0.03 %)

Overflow GRC

829.02

Cell area (μm²)

refine_opt — Constraints

early timing endpoint 1개 (clk_r_REG1_S2/D) — CTS 단계에서 해소, design rule 위반 0.

report_constraints -all_violators · after refine_opt

```
****************************************

   late_timing
   -----------

Endpoint                        Path Delay     Path Required       CRP    Slack Group    Scenario
-------------------------------------------------------------------------------------------------
No paths.

   early_timing
   -----------

Endpoint                        Path Delay     Path Required       CRP    Slack Group    Scenario
-------------------------------------------------------------------------------------------------
clk_r_REG1_S2/D (DFFARX1_HVT)      0.32 f         0.32           0.00    -0.00   clk      normal_tt1p05vn40c

   Mode: normal Corner: tt0p78v125c
   Scenario: normal_tt0p78v125c
  ---------------------------------------------------------------------------
   Number of max_transition violation(s): 0

   Mode: normal Corner: tt1p05vn40c
   Scenario: normal_tt1p05vn40c
  ---------------------------------------------------------------------------
   Number of max_transition violation(s): 0

   Mode: normal Corner: tt0p78v125c
   Scenario: normal_tt0p78v125c
  ---------------------------------------------------------------------------
   Number of max_capacitance violation(s): 0

   Mode: normal Corner: tt1p05vn40c
   Scenario: normal_tt1p05vn40c
  ---------------------------------------------------------------------------
   Number of max_capacitance violation(s): 0


   Mode: normal Corner: tt0p78v125c
   Scenario: normal_tt0p78v125c
  ---------------------------------------------------------------------------
   Number of min_capacitance violation(s): 0

   Mode: normal Corner: tt1p05vn40c
   Scenario: normal_tt1p05vn40c
  ---------------------------------------------------------------------------
   Number of min_capacitance violation(s): 0

   Total number of violation(s): 0
```

1

Hold endpoint

tt1p05vn40c

0

DRC

trans · cap

refine_opt — Vth Ratio

LVT 13 / 261 cell (4.98 %) — spec < 20 % 만족.

report_threshold_voltage_group · after refine_opt

```
****************************************
Report : threshold_voltage_group
Design : TOP
Version: T-2022.03-SP5
Date   : Mon Oct 12 11:09:25 2026
****************************************

Cell Count Report
-----------------------------------------------------------------------------------------------------------------------------------------------------------------

Group Names                     All (%)       Repeaters (%)   Combinational (%)      Registers (%)     Sequentials (%)   Clock_network (%)          Macro (%)   Physical_only (%)
-----------------------------------------------------------------------------------------------------------------------------------------------------------------
saed32cell_hvt              76  (29.12%)       54  (20.69%)       19   (7.28%)        3   (1.15%)        0   (0.00%)        0   (0.00%)        0   (0.00%)        0   (0.00%)
saed32cell_lvt              13   (4.98%)        3   (1.15%)        1   (0.38%)        9   (3.45%)        0   (0.00%)        0   (0.00%)        0   (0.00%)        0   (0.00%)  L
saed32cell_svt             172  (65.90%)       35  (13.41%)       89  (34.10%)       48  (18.39%)        0   (0.00%)        0   (0.00%)        0   (0.00%)        0   (0.00%)
-----------------------------------------------------------------------------------------------------------------------------------------------------------------
Total                      261 (100.00%)       92  (35.25%)      109  (41.76%)       60  (22.99%)        0   (0.00%)        0   (0.00%)        0   (0.00%)        0   (0.00%)
Low vth groups              13   (4.98%)        3   (1.15%)        1   (0.38%)        9   (3.45%)        0   (0.00%)        0   (0.00%)        0   (0.00%)        0   (0.00%)
-----------------------------------------------------------------------------------------------------------------------------------------------------------------


Cell Area Report
-----------------------------------------------------------------------------------------------------------------------------------------------------------------

Group Names                     All (%)       Repeaters (%)   Combinational (%)      Registers (%)     Sequentials (%)   Clock_network (%)          Macro (%)   Physical_only (%)
-----------------------------------------------------------------------------------------------------------------------------------------------------------------
saed32cell_hvt          137.75  (16.62%)     73.96   (8.92%)     42.44   (5.12%)     21.35   (2.58%)      0.00   (0.00%)      0.00   (0.00%)      0.00   (0.00%)      0.00   (0.00%)
saed32cell_lvt           70.91   (8.55%)      4.83   (0.58%)      2.03   (0.25%)     64.04   (7.73%)      0.00   (0.00%)      0.00   (0.00%)      0.00   (0.00%)      0.00   (0.00%)  L
saed32cell_svt          620.37  (74.83%)     56.67   (6.84%)    222.12  (26.79%)    341.57  (41.20%)      0.00   (0.00%)      0.00   (0.00%)      0.00   (0.00%)      0.00   (0.00%)
-----------------------------------------------------------------------------------------------------------------------------------------------------------------
Total                   829.02 (100.00%)    135.46  (16.34%)    266.60  (32.16%)    426.96  (51.50%)      0.00   (0.00%)      0.00   (0.00%)      0.00   (0.00%)      0.00   (0.00%)
Low vth groups           70.91   (8.55%)      4.83   (0.58%)      2.03   (0.25%)     64.04   (7.73%)      0.00   (0.00%)      0.00   (0.00%)      0.00   (0.00%)      0.00   (0.00%)
-----------------------------------------------------------------------------------------------------------------------------------------------------------------
```

4.98 %
LVT
13 / 261

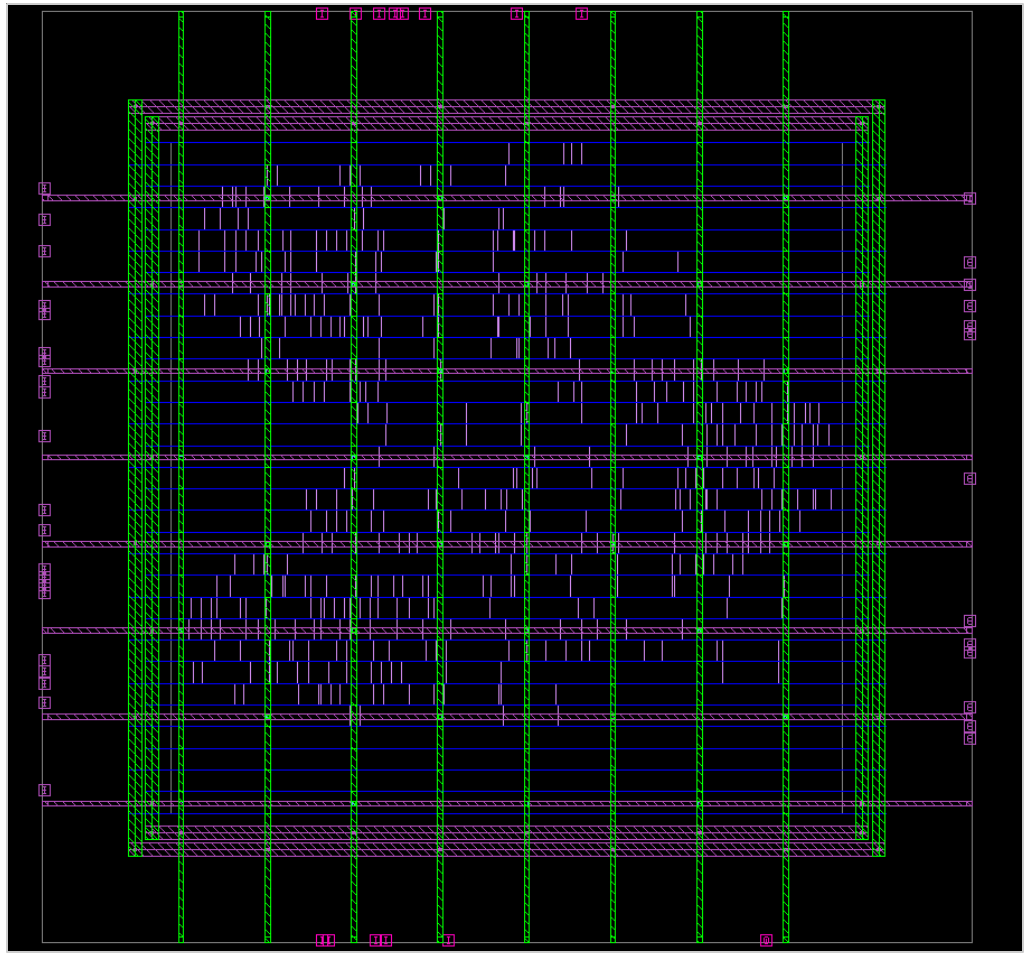
29.12 %
HVT
76

65.90 %
SVT
172

Place Result

place 완료 layout — utilization 0.4488 → 0.3077.

layout · after place



0.3077

Utilization

829.02

Cell area (μm²)

floorplan 1,209.22

04

SECTION

CTS

balanced clock tree를 build · route하고 propagated clock 기준으로 setup · hold 최적화.

정태웅 · 청주대학교

cts.tcl (1/3)

pre-CTS check → 전역 옵션 → 기존 tree · CCD offset 정리.

cts.tcl · 0–2

```
#--------------------------------------------------------------------------
# 0. Open the placed design and check CTS readiness
#--------------------------------------------------------------------------
open_lib   nlib/place.nlib/
open_block /home/AI23/all_lab_student/tools_TOP/icc2/nlib/place.nlib:TOP.design
link_block
check_design -checks pre_clock_tree_stage

#--------------------------------------------------------------------------
# 1. Global CTS / optimization options
#--------------------------------------------------------------------------
# Promote clock nets to NDR only where congestion allows
set_app_options -list {cts.optimize.enable_congestion_aware_ndr_promotion true}

# Fix hold in both scenarios
set_scenario_status {normal_tt0p78v125c normal_tt1p05vn40c} -hold true

# Remove common-path pessimism on reconvergent clock paths (CRPR)
set_app_options -list {time.remove_clock_reconvergence_pessimism true}

#--------------------------------------------------------------------------
# 2. Pre-CTS setup
#--------------------------------------------------------------------------
# Disallow full-adder cells (dont_use from synthesis)
set_lib_cell_purpose -include none [get_lib_cells */FADD*]

# Start from a clean clock net (drop any previous tree)
remove_clock_trees -clocks [get_clocks clk]

# Check CCD, local skew status and clear CCD offsets left by place_opt (balanced tree, no useful skew)
report_app_options *enable_ccd*
report_app_options *local_skew*
remove_clock_balance_points -clock clk -scenarios {normal_tt0p78v125c normal_tt1p05vn40c}
report_clock_balance_points
```

Option
congestion-aware NDR promotion, 두 scenario hold fix on, CRPR on.

Pre-CTS
FADD* cell 사용 제외, 기존 clock tree 제거.

CCD 정리
clock balance point 제거 — useful skew 미사용.

cts.tcl (2/3)

CTS target · clock design rule · clock cell · NDR 지정 후 build → route clock.

cts.tcl · 3–5

```
#--------------------------------------------------------------------------------
# 3. CTS targets and design rules
#--------------------------------------------------------------------------------
# Targets (best effort): latency 0.6ns, skew 0.05ns on both corners
set_clock_tree_options -clocks [get_clocks clk] -corners {tt0p78v125c tt1p05vn40c} \
        -target_latency 0.6 -target_skew 0.05

# Clock path transition limit 0.12ns and clock net max fanout 12
set_max_transition 0.12 -clock_path [get_clocks clk] -scenarios {normal_tt0p78v125c normal_tt1p05vn40c}
set_app_options -name cts.common.max_fanout -value 12

# Relax design / port max capacitance to 10 on both scenarios
set_max_capacitance 10 [current_design] -scenarios {normal_tt0p78v125c normal_tt1p05vn40c}
set_max_capacitance 10 [all_inputs]     -scenarios {normal_tt0p78v125c normal_tt1p05vn40c}
set_max_capacitance 10 [all_outputs]    -scenarios {normal_tt0p78v125c normal_tt1p05vn40c}

# Keep LVT/HVT buffers and inverters out of the clock tree (RVT only)
set_lib_cell_purpose -exclude cts [get_lib_cells "*/*BUF*_LVT */*INV*_LVT */*BUF*_HVT */*INV*_HVT"]

#--------------------------------------------------------------------------------
# 4. Clock NDR : root / internal nets on M3-M4, sink nets with default rule
#--------------------------------------------------------------------------------
create_routing_rule 2xS_2xW_RULE \
        -width    {M1 0.1 M2 0.112 M3 0.112 M4 0.112} \
        -spacings {M1 0.1 M2 0.112 M3 0.112 M4 0.112}

set_clock_routing_rules -clocks [get_clocks clk] -net_type root     -rules 2xS_2xW_RULE -min_routing_layer M3 -max_routing_layer M4
set_clock_routing_rules -clocks [get_clocks clk] -net_type internal -rules 2xS_2xW_RULE -min_routing_layer M3 -max_routing_layer M4
set_clock_routing_rules -clocks [get_clocks clk] -net_type sink     -default_rule       -min_routing_layer M1 -max_routing_layer M4

#--------------------------------------------------------------------------------
# 5. Build and route the clock tree
#--------------------------------------------------------------------------------
clock_opt -from build_clock -to route_clock
```

Target
latency 0.6 · skew 0.05 (best effort).

Rule
clock max_transition 0.12 · max_fanout 12.

Clock cell
LVT · HVT buffer / inverter 제외 → RVT만.

NDR
root · internal 2xS_2xW on M3 – M4, sink default.

cts.tcl (3/3)

post-CTS clock 제약 적용 후 final_opto.

cts.tcl · 6–7

```
#--------------------------------------------------------------------------
# 6. Post-CTS clock constraints
#    propagated clock, source latency 0.15, uncertainty setup 0.02 / hold 0.03
#--------------------------------------------------------------------------
foreach scn {normal_tt0p78v125c normal_tt1p05vn40c} {
        current_scenario $scn
        set_propagated_clock   [get_clocks clk]
        set_clock_latency -source 0.15 [get_clocks clk]
        set_clock_uncertainty -setup 0.02 [get_clocks clk]
        set_clock_uncertainty -hold  0.03 [get_clocks clk]
}

#--------------------------------------------------------------------------
# 7. Final optimization (setup + hold) and reports
#--------------------------------------------------------------------------
clock_opt -from final_opto
report_qor -summary -significant_digits 5
report_qor
report_clock_qor
report_utilization

report_qor -summary -significant_digits 5 > reports/cts/report_qor_summary_after_cts.txt
report_qor > reports/cts/report_qor_after_cts.txt
report_clock_qor > reports/cts/report_clock_qor_after_cts.txt
report_utilization > reports/cts/report_utilization_after_cts.txt

save_lib -as nlib/cts.nlib
```

Propagated
source latency 0.15 — spec 외부 clock generator delay.

Uncertainty
setup 0.02 = jitter 10 ps + setup margin 10 ps, hold 0.03 = spec minimum.

Final opt
clock_opt -from final_opto로 setup · hold 최적화.

Pre-CTS Check

check_design pre_clock_tree_stage — error 0, 앞 단계에 없던 warning만 설명.

check_design · pre_clock_tree_stage

0 ERRORS

```
****************************************
 Report : check_design
 Options: { pre_clock_tree_stage }
 Design : TOP
 Version: T-2022.03-SP5
 Date   : Mon Oct 12 11:11:27 2026
****************************************

Running mega-check 'pre_clock_tree_stage':
    Running atomic-check 'design_mismatch'
    Running atomic-check 'scan_chain'
    Running atomic-check 'mv_design'
    Running atomic-check 'legality'
    Running atomic-check 'timing'
    Running atomic-check 'clock_trees'
    Running atomic-check 'hier_pre_clock_tree'

  *** EMS Message summary ***
  ------------------------------------------------------------------------------------------------
  Rule         Type   Count      Message
  ------------------------------------------------------------------------------------------------
  DFT-011      Info   1          The design has no scan chain defined in the scandef.
  TCK-001      Warn   60         The reported endpoint '%endpoint' is unconstrained. Reason: '%re...
  TCK-012      Warn   2          The input port '%port' has no clock_relative delay specified. Mo...
  ------------------------------------------------------------------------------------------------
  Total 63 EMS messages : 0 errors, 62 warnings, 1 info.
  ------------------------------------------------------------------------------------------------

  *** Non-EMS message summary ***
  ------------------------------------------------------------------------------------------------
  Rule         Type   Count      Message
  ------------------------------------------------------------------------------------------------
  CTS-101      Info   1          %s will work on the following scenarios.
  CTS-107      Info   1          %s will work on all clocks in active scenarios, including %d mas...
  CTS-973      Info   1          The value of option cts.compile.enable_cell_relocation has been ...
  NEX-001      Warn   6          Technology layer '%s' setting '%s' is not valid
  PDC-003      Warn   14         Routing direction of metal layer %s is neither "horizontal" nor ...
  PVT-032      Info   2          Corner %s: no PVT mismatches.
  ------------------------------------------------------------------------------------------------
  Total 25 non-EMS messages : 0 errors, 20 warnings, 5 info.
```

PDC-003 (14)

routing direction 미지정 metal layer → 해당 layer는 PDC check에서 제외.

Clock QoR — After CTS

clk 60 sink · 13 level · repeater 13, trans · cap DRC 0.

report_clock_qor · after CTS

```
============================================================ Summary Table for Corner tt0p78v125c ============================================
Clock /                                 Attrs     Sinks Levels    Clock     Clock     Clock       Max    Global  Trans DRC  Cap DRC      Wire
Skew Group                                                       Repeater  Repeater  Stdcell   Latency      Skew     Count     Count    Length
                                                                    Count      Area      Area
------------------------------------------------------------------------------------------------------------------------------------------
### Mode: normal, Scenario: normal_tt0p78v125c
clk                                      M,D        60     13       13     27.96     69.89      0.61      0.01         0         0    656.71
------------------------------------------------------------------------------------------------------------------------------------------
All Clocks                                          60     13       13     27.96     69.89      0.61      0.01         0         0    656.71


Warning: Please use -largest / -smallest / -all switches with -show_verbose_paths / -show_paths to report the clock paths. (CTS-956)
================================================
==== Summary Reporting for Corner tt1p05vn40c ====
================================================

============================================================ Summary Table for Corner tt1p05vn40c ============================================
Clock /                                 Attrs     Sinks Levels    Clock     Clock     Clock       Max    Global  Trans DRC  Cap DRC      Wire
Skew Group                                                       Repeater  Repeater  Stdcell   Latency      Skew     Count     Count    Length
                                                                    Count      Area      Area
------------------------------------------------------------------------------------------------------------------------------------------
### Mode: normal, Scenario: normal_tt1p05vn40c
clk                                      M,D        60     13       13     27.96     69.89      0.34      0.01         0         0    656.71
------------------------------------------------------------------------------------------------------------------------------------------
All Clocks                                          60     13       13     27.96     69.89      0.34      0.01         0         0    656.71
```

0.61 / 0.01

tt0p78v125c

latency / skew

0.34 / 0.01

tt1p05vn40c

latency / skew

0 / 0

Trans / Cap DRC

Target 대비

skew 0.05 충족, slow corner latency 0.61은 target 0.6 소폭 초과 (best effort).

QoR — After CTS

propagated clock 기준 setup · hold 위반 0 — place 잔여 hold 해소.

report_qor -summary · after CTS

0 NVE

```
icc2_shell> report_qor -summary -significant_digits 5
****************************************
Report : qor
         -summary
Design : TOP
Version: T-2022.03-SP5
Date   : Mon Oct 12 11:14:26 2026
****************************************
Information: Timer using 'CRPR'. (TIM-050)

Timing
--------------------------------------------------------------------------------
Context                                 WNS             TNS             NVE
--------------------------------------------------------------------------------
normal_tt0p78v125c (Setup)          0.00972         0.00000               0
normal_tt1p05vn40c (Setup)          0.20892         0.00000               0
Design             (Setup)          0.00972         0.00000               0

normal_tt0p78v125c (Hold)           0.00121         0.00000               0
normal_tt1p05vn40c (Hold)           0.01039         0.00000               0
Design             (Hold)           0.00121         0.00000               0
--------------------------------------------------------------------------------

Miscellaneous
--------------------------------------------------------------------------------
Cell Area (netlist):                          885.18355
Cell Area (netlist and physical only):        885.18400
Nets with DRC Violations:        0
```

+0.00972

Setup WNS

tt0p78v125c

+0.00121

Hold WNS

tt0p78v125c

885.18

Cell area (μm²)

place 829.02

Constraints — After CTS

report_constraints — 위반 0.

report_constraints -all_violators · after CTS **0 VIOLATIONS**

```
Mode: normal Corner: tt0p78v125c
Scenario: normal_tt0p78v125c
max_transition
                          Required       Actual
Net                      Transition     Transition       Slack  Violation
-------------------------------------------------------------------------
HFSNET_13                   0.12           0.12           0.00  (MET)

-------------------------------------------------------------------------
Number of max_transition violation(s): 0

Mode: normal Corner: tt1p05vn40c
Scenario: normal_tt1p05vn40c
max_transition
                          Required       Actual
Net                      Transition     Transition       Slack  Violation
-------------------------------------------------------------------------
imax_1/n34                  0.05           0.03           0.02  (MET)

-------------------------------------------------------------------------
Number of max_transition violation(s): 0

Mode: normal Corner: tt0p78v125c
Scenario: normal_tt0p78v125c
max_capacitance
                          Required       Actual
Net                      Capacitance    Capacitance      Slack  Violation
-------------------------------------------------------------------------
HFSNET_9                   10.00           9.99           0.01  (MET)

-------------------------------------------------------------------------
Number of max_capacitance violation(s): 0

Mode: normal Corner: tt1p05vn40c
Scenario: normal_tt1p05vn40c
max_capacitance
                          Required       Actual
Net                      Capacitance    Capacitance      Slack  Violation
-------------------------------------------------------------------------
HFSNET_9                   10.00           9.79           0.21  (MET)

-------------------------------------------------------------------------
Number of max_capacitance violation(s): 0


Mode: normal Corner: tt0p78v125c
Scenario: normal_tt0p78v125c
-------------------------------------------------------------------------
Number of min_capacitance violation(s): 0

Mode: normal Corner: tt1p05vn40c
Scenario: normal_tt1p05vn40c
-------------------------------------------------------------------------
Number of min_capacitance violation(s): 0

Total number of violation(s): 0
```

0

Total violations

Worst 항목

max_transition HFSNET_13 0.12 / 0.12, max_cap HFSNET_9 9.99 / 10.00 (tt0p78v125c) — MET.

Vth Ratio — After CTS

LVT 3 / 273 cell (1.10 %) — clock network 23 cell은 모두 SVT.

report_threshold_voltage_groups · after CTS

```
icc2_shell> report_threshold_voltage_groups
****************************************
Report : threshold_voltage_group
Design : TOP
Version: T-2022.03-SP5
Date   : Mon Oct 12 11:17:21 2026
****************************************

Cell Count Report
```

Group Names	All (%)	Repeaters (%)	Combinational (%)	Registers (%)	Sequentials (%)	Clock_network (%)	Macro (%)	Physical_only (%)	
saed32cell_hvt	83 (30.40%)	59 (21.61%)	21 (7.69%)	3 (1.10%)	0 (0.00%)	0 (0.00%)	0 (0.00%)	0 (0.00%)	
saed32cell_lvt	3 (1.10%)	1 (0.37%)	2 (0.73%)	0 (0.00%)	0 (0.00%)	0 (0.00%)	0 (0.00%)	0 (0.00%)	L
saed32cell_svt	187 (68.50%)	24 (8.79%)	83 (30.40%)	57 (20.88%)	0 (0.00%)	23 (8.42%)	0 (0.00%)	0 (0.00%)	
Total	273 (100.00%)	84 (30.77%)	106 (38.83%)	60 (21.98%)	0 (0.00%)	23 (8.42%)	0 (0.00%)	0 (0.00%)	
Low vth groups	3 (1.10%)	1 (0.37%)	2 (0.73%)	0 (0.00%)	0 (0.00%)	0 (0.00%)	0 (0.00%)	0 (0.00%)	

Cell Area Report

Group Names	All (%)	Repeaters (%)	Combinational (%)	Registers (%)	Sequentials (%)	Clock_network (%)	Macro (%)	Physical_only (%)	
saed32cell_hvt	165.96 (18.75%)	93.78 (10.59%)	50.83 (5.74%)	21.35 (2.41%)	0.00 (0.00%)	0.00 (0.00%)	0.00 (0.00%)	0.00 (0.00%)	
saed32cell_lvt	7.88 (0.89%)	1.27 (0.14%)	6.61 (0.75%)	0.00 (0.00%)	0.00 (0.00%)	0.00 (0.00%)	0.00 (0.00%)	0.00 (0.00%)	L
saed32cell_svt	711.35 (80.36%)	30.75 (3.47%)	205.09 (23.17%)	405.61 (45.82%)	0.00 (0.00%)	69.89 (7.90%)	0.00 (0.00%)	0.00 (0.00%)	
Total	885.18 (100.00%)	125.80 (14.21%)	262.53 (29.66%)	426.96 (48.23%)	0.00 (0.00%)	69.89 (7.90%)	0.00 (0.00%)	0.00 (0.00%)	
Low vth groups	7.88 (0.89%)	1.27 (0.14%)	6.61 (0.75%)	0.00 (0.00%)	0.00 (0.00%)	0.00 (0.00%)	0.00 (0.00%)	0.00 (0.00%)	

1.10 %
LVT
3 / 273

30.40 %
HVT
83

68.50 %
SVT
187

Utilization — After CTS

report_utilization — 0.3285, cell area 885.18.

report_utilization · after CTS

```
icc2_shell> report_utilization
****************************************
Report : report_utilization
Design : TOP
Version: T-2022.03-SP5
Date   : Mon Oct 12 11:19:36 2026
****************************************
Utilization Ratio:                      0.3285
Utilization options:
 - Area calculation based on:           site_row of block TOP
 - Categories of objects excluded:      hard_macros macro_keepouts soft_macros io_cells hard_blockages
Total Area:                             2694.4347
Total Capacity Area:                    2694.4347
Total Area of cells:                    885.1836
Area of excluded objects:
 - hard_macros          :               0.0000
 - macro_keepouts       :               0.0000
 - soft_macros          :               0.0000
 - io_cells             :               0.0000
 - hard_blockages       :               0.0000

Utilization of site-rows with:
 - Site 'unit':                         0.3285

0.3285
```

0.3285

Utilization

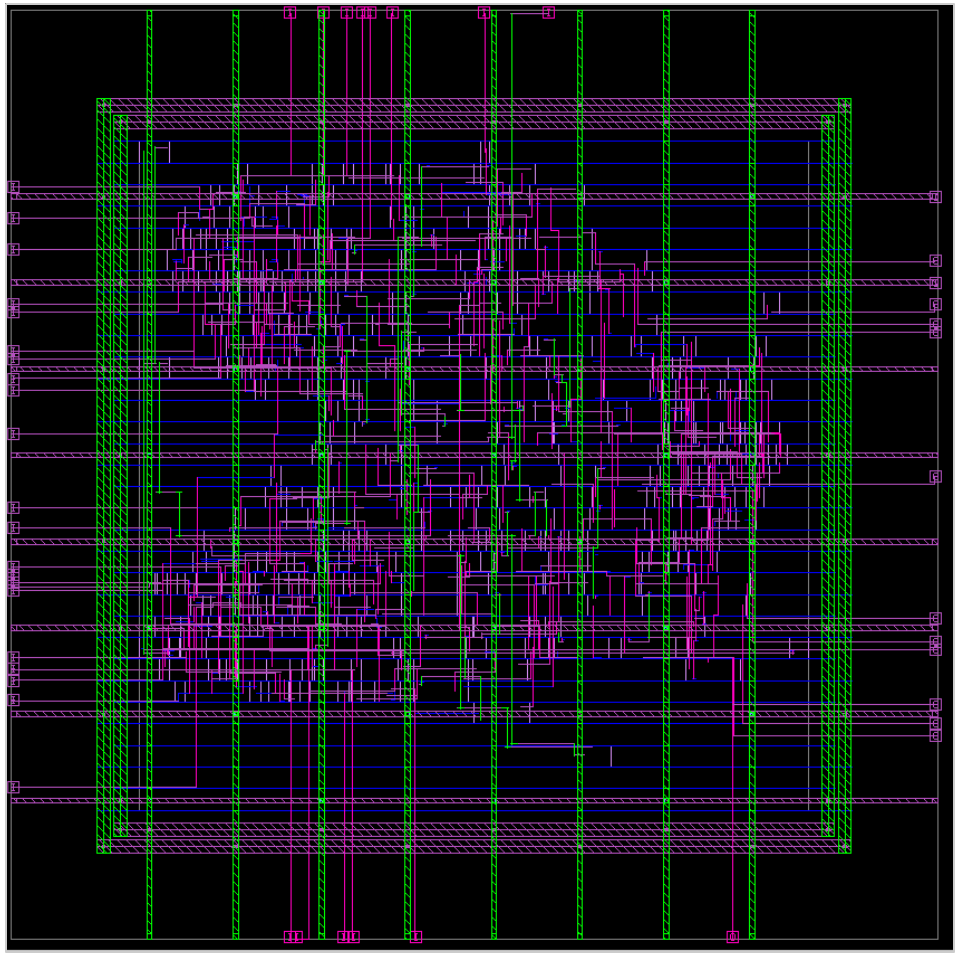
885.18

Cell area (μm²)

CTS Result

clock tree 배선 완료 layout.

layout · after CTS



0.3285

Utilization

13

Clock repeaters

27.96 μm²

05

SECTION

Route

routability 점검 후 route_auto → route_opt 진행, 도중 발생한 zroute error 해결.

정태웅 · 청주대학교

route.tcl (1/3)

pre-route check → redundant via · route 옵션 → route_auto.

route.tcl · 0–3

```
#-------------------------------------------------------------------------------
# 0. Open the CTS design and run pre-route checks
#-------------------------------------------------------------------------------
open_lib   nlib/cts.nlib/
open_block /home/AI23/all_lab_student/tools_TOP/icc2/nlib/cts.nlib:TOP.design
link_block

check_routability
check_design -checks pre_route_stage

#-------------------------------------------------------------------------------
# 1. Redundant via insertion (after detail route)
#-------------------------------------------------------------------------------
set_app_options -name route.common.post_detail_route_redundant_via_insertion -value medium;
add_redundant_vias -list_only true      ;# list mapping list

#-------------------------------------------------------------------------------
# 2. Route options : timing-driven, crosstalk-driven only in track assignment
#-------------------------------------------------------------------------------
set_app_options -list {
        route.global.timing_driven     true
        route.global.crosstalk_driven  false
        route.track.timing_driven      true
        route.track.crosstalk_driven   true
        route.detail.timing_driven     true
}

#-------------------------------------------------------------------------------
# 3. Global route -> track assignment -> detail route
#-------------------------------------------------------------------------------
route_auto
check_routes
```

Redundant via
post detail route redundant via insertion medium.

Option
global · track · detail timing-driven, crosstalk-driven은 track만.

route_auto
global → track assignment → detail route.

route.tcl (2/3)

SI analysis on → route_opt → post-route check.

route.tcl · 4–5

```
#-------------------------------------------------------------------------
# 4. Post-route optimization with SI analysis
#-------------------------------------------------------------------------
set_app_options -list {time.si_enable_analysis true}

route_opt
report_qor -summary -significant_digits 5
report_congestion

#-------------------------------------------------------------------------
# 5. Post-route checks and save
#-------------------------------------------------------------------------
check_routes
check_legality
check_pg_drc
check_pg_connectivity

save_lib -as nlib/route.nlib
```

route_opt
time.si_enable_analysis true 후 post-route 최적화.

Check
check_routes · check_legality · check_pg_drc · check_pg_connectivity.

Pre-route Check

check_design pre_route_stage — error 0, 앞 단계에 없던 warning만 설명.

check_design · pre_route_stage **0 ERRORS**

```
****************************************
 Report : check_design
 Options: { pre_route_stage }
 Design : TOP
 Version: T-2022.03-SP5
 Date   : Mon Oct 12 11:23:11 2026
****************************************

Running mega-check 'pre_route_stage':
    Running atomic-check 'design_mismatch'
    Running atomic-check 'scan_chain'
    Running atomic-check 'mv_design'
    Running atomic-check 'timing'
    Running atomic-check 'routability'
    Running atomic-check 'hier_pre_route'

  *** EMS Message summary ***
  ------------------------------------------------------------------------------------------------
  Rule         Type   Count      Message
  ------------------------------------------------------------------------------------------------
  DFT-011      Info   1          The design has no scan chain defined in the scandef.
  TCK-001      Warn   60         The reported endpoint '%endpoint' is unconstrained. Reason: '%re...
  TCK-012      Warn   2          The input port '%port' has no clock_relative delay specified. Mo...
  ------------------------------------------------------------------------------------------------
  Total 63 EMS messages : 0 errors, 62 warnings, 1 info.
  ------------------------------------------------------------------------------------------------

  *** Non-EMS message summary ***
  ------------------------------------------------------------------------------------------------
  Rule         Type   Count      Message
  ------------------------------------------------------------------------------------------------
  NEX-001      Warn   6          Technology layer '%s' setting '%s' is not valid
  PVT-032      Info   2          Corner %s: no PVT mismatches.
  ZRT-022      Warn   1          Cannot find a default contact code for layer %s.
  ZRT-025      Warn   6          Layer %s does not have a preferred direction, assigning to %s.
  ZRT-619      Info   1          Via ladder engine would be activated for pattern must join conne...
  ZRT-703      Info   1          Option route.detail.force_end_on_preferred_grid will be ignored ...
  ZRT-706      Info   1          When applicable layer based tapering will taper up to %.2f in di...
  ZRT-707      Info   1          When applicable Min-max layer allow_pin_connection mode will all...
  ZRT-718      Info   1          When applicable Min-max layer allow_pin_connection mode will all...
  ------------------------------------------------------------------------------------------------
  Total 20 non-EMS messages : 0 errors, 13 warnings, 7 info.
  ------------------------------------------------------------------------------------------------
```

ZRT-025 (6)
preferred direction 없는 layer(M5 – M9 · MRDL)에 방향 자동 지정 — routing 범위 M1 – M4 밖이라 무관.

ZRT-022 (1)
layer CO에 default contact code 없음 — non-routing layer라 무관 (man).

check_routability (1/3)

port · PG · overlap · min-grid · boundary · blocked port/net — 문제 없음.

check_routability · 1

```
==========================================
==  Check for design                    ==
==========================================

>>> No net contains a large number of ports

>>> No port contains a large number of pins


================================
==  Check for PG DPT on Track ==
================================
>>>>>> Number of PG rails cross even number of track: 0

==========================================
==  Check for PG PreRoute setting       ==
==========================================
 No number_of_secondary_pg_pin_connections setting and skip checking

==========================================
==  Check for overlap of standard cells ==
==========================================

>>>>>> No overlap of standard cells found

==========================================
==     Check for min-grid violations    ==
==========================================

>>>>>> No min-grid violations found


==========================================
==    Check for out-of-boundary ports   ==
==========================================

>>>>>> No out-of-boundary error found

==========================================
==        Check for blocked ports       ==
==========================================

>>>>>> Port might be blocked by layer constraints - min/max and freeze layer settings

>>>>>> Port might be blocked by check port access

>>>>>> No blocked ports found

>>>>>> Net blocked by layer constraints - min/max and freeze layer settings

>>>>>> No blocked nets found
```

check_routability (2/3)

shielding · via cut blockage · custom via · NDR 설정 — 문제 없음.

check_routability · 2

```
==========================================
==     Check for shielding violations     ==
==========================================

>>>>>> The following Non-PG net has a shape with shape_use shield_route

>>> No non-PG net has a shape with shape_use shield_route.

>>>>>> The following PG net has shape with shape_use detail_route and shape length is too long.

>>> No PG net has shape with shape_use detail_route and shape length is too long.

>>>>>> The following nets with shield non-default rule has no internal data representation in Zroute.

>>> No nets with shield non-default rule has no internal data representation in Zroute.

==========================================
==     Check for via cut blockage violation   ==
==========================================

>>>>>> The following via cut blockages are wrongly treated as real vias.

>>> No via cut blockage is wrongly treated as real via.

==========================================
==     Check for custom via definition    ==
==========================================

>>> No custom via definition has too many cuts

==========================================
==          Check for via master          ==
==========================================

>>> The total number of via master definition is 299

==========================================
==     Check non-default rule setting        ==
==========================================

>>> No NDR width is larger than signalRouteMaxWidth

>>> No NDR shield width is larger than signalRouteMaxWidth
```

check_routability (3/3)

중복 PG shape · non-routing layer pin · over-promoted net — 문제 없음.

check_routability · 3 PASS

```
==========================================================
==      Check redundant duplicated PG shapes            ==
==========================================================

>>> No redundant duplicated PG shapes overlap with each other.

==========================================================
==      Check library cell has non-routing layer pins   ==
==========================================================

>>> No cell has non-routing layer pins to be connected.

==========================================
==      Check over promoted nets        ==
==========================================

>>> No over promoted nets.

End of check_routability
```

route_auto — check_routes

Issue 해결 — 수정된 ref library로 재실행, route_auto 후 DRC 0.

check_routes · after route_auto

RESOLVED

```
Verify Summary:

Total number of nets = 311, of which 0 are not extracted
Total number of open nets = 0, of which 0 are frozen
Total number of excluded ports = 0 ports of 0 unplaced cells connected to 0 nets
                                 0 ports without pins of 0 cells connected to 0 nets
                                 0 ports of 0 cover cells connected to 0 non-pg nets
Total number of DRCs = 0
Total number of antenna violations = no antenna rules defined
Total number of tie to rail violations = not checked
Total number of tie to rail directly violations = not checked
```

0
DRCs

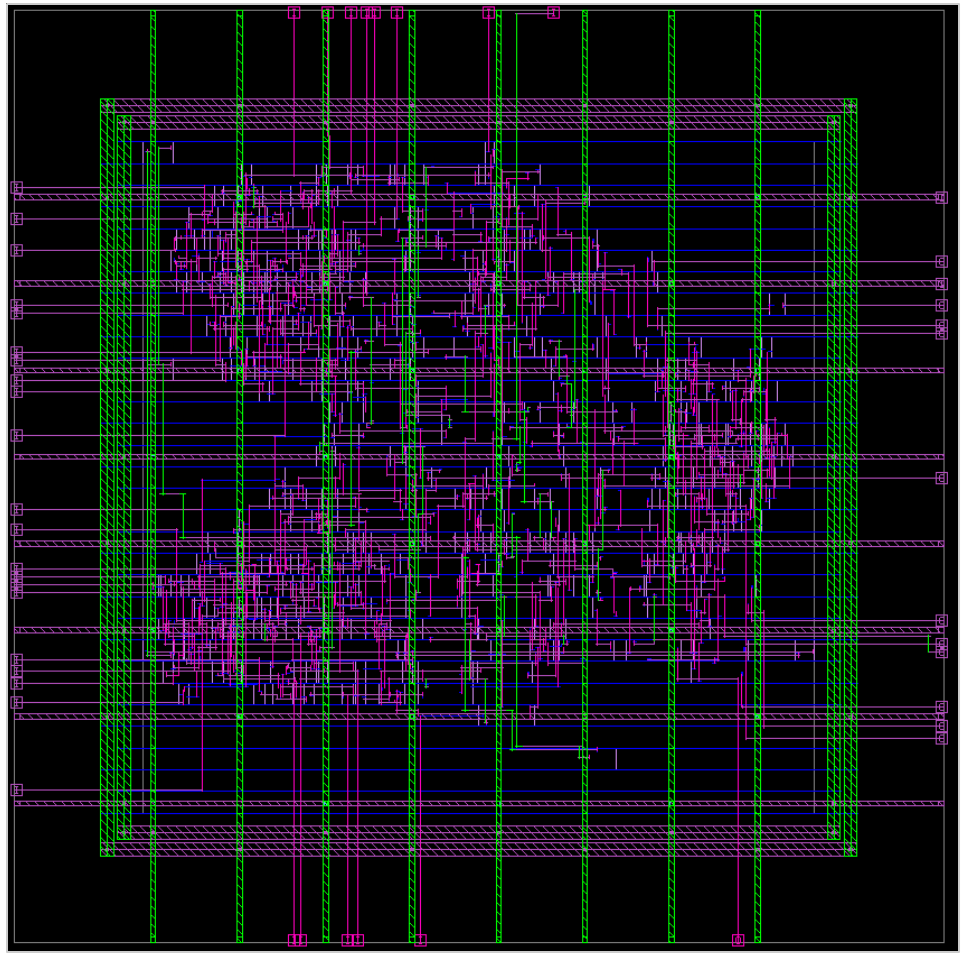
0
Open nets

311
Nets

route_auto Layout

global → track → detail route 완료 layout, M1 – M4 범위 배선.

layout · after route_auto

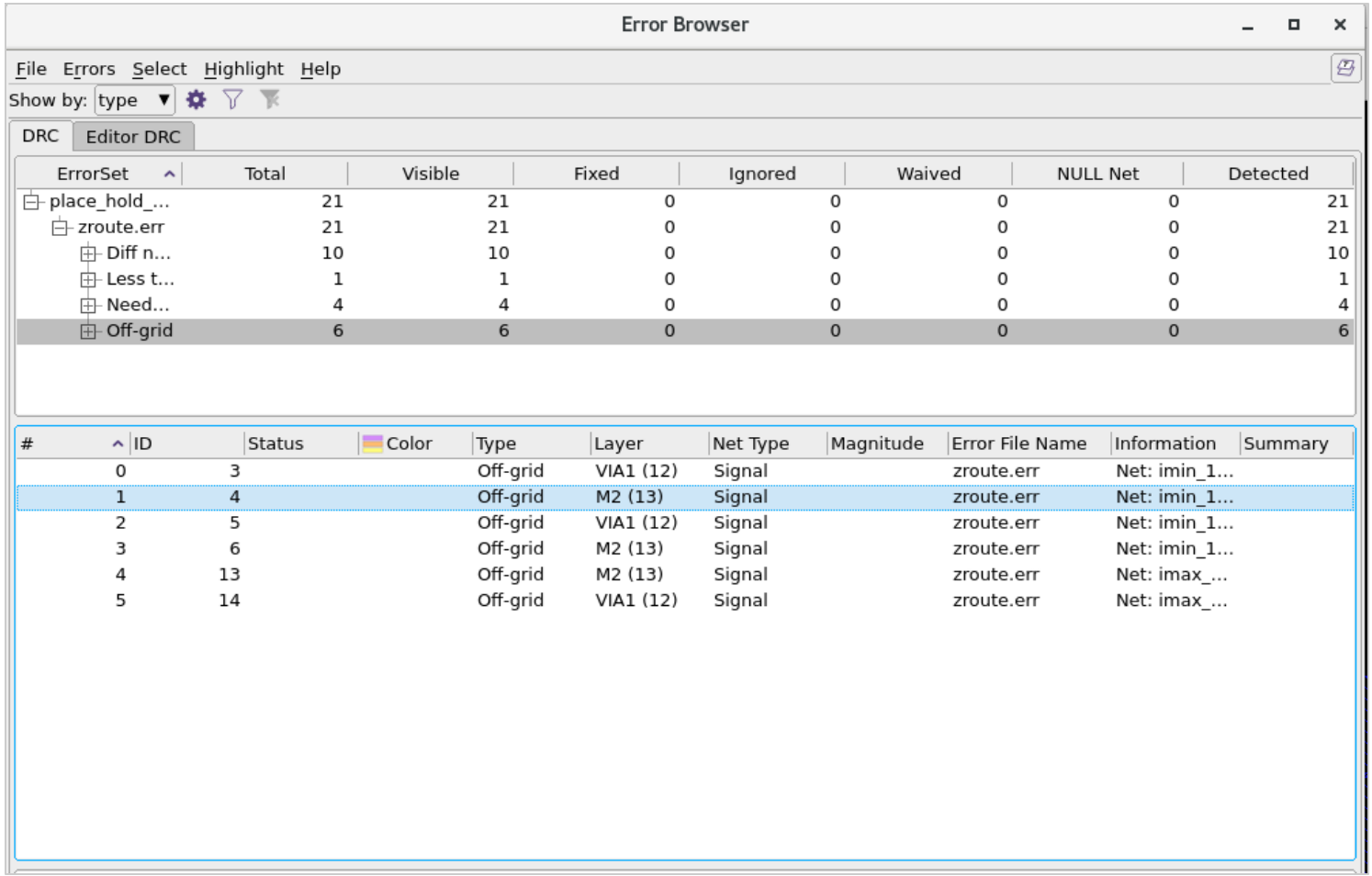


zroute Error

Issue — route 후 zroute error 21건, OR / NOR gate에서 발생.

Error Browser · zroute.err

VIOLATED



21

zroute.err 전체

6

Off-grid (예시)

VIA1 · M2

무엇

route 후 zroute error 21건 — 목록은 그중 Off-grid 6건.

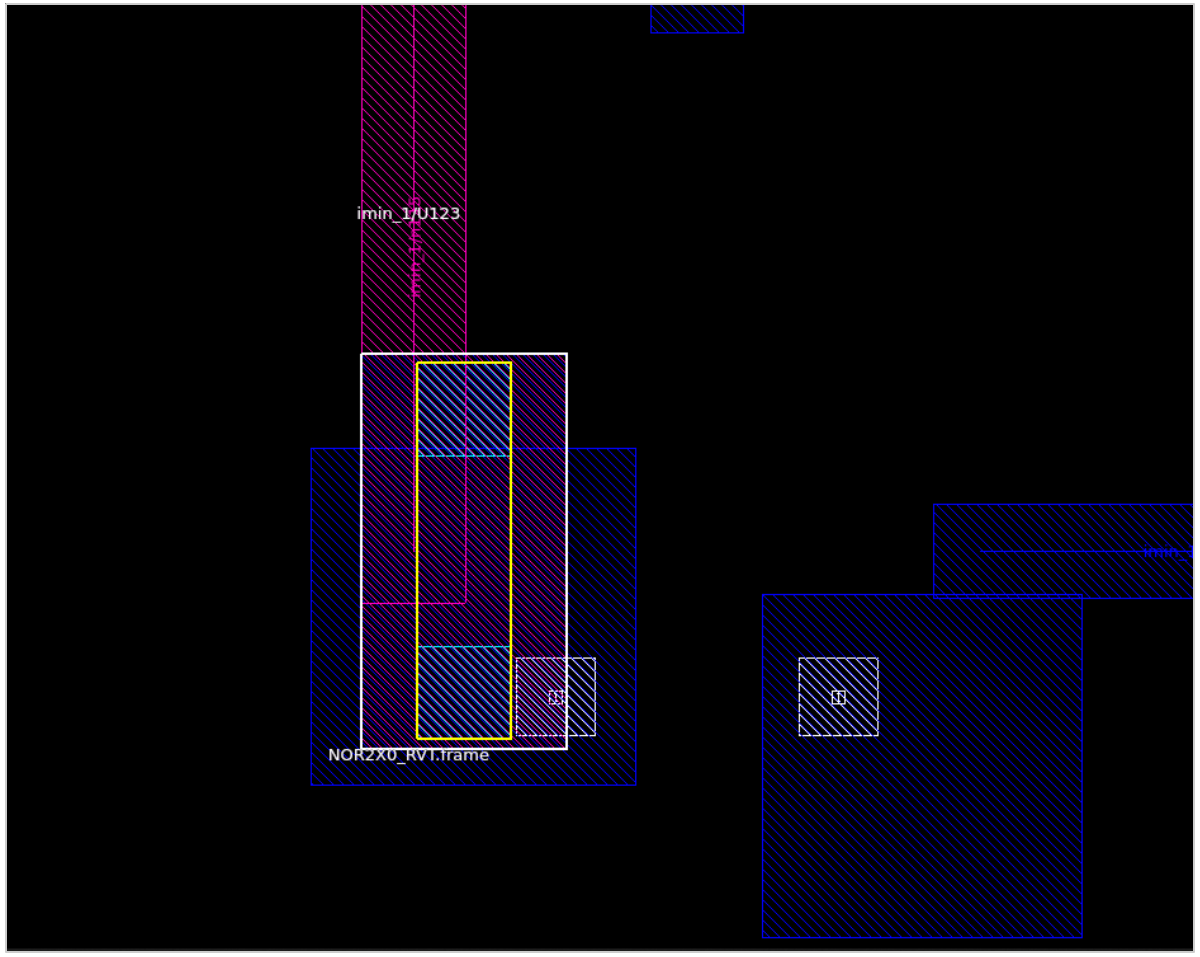
위치

error 지점이 OR / NOR gate cell에 집중.

zroute Error — 원인

Issue — Off-grid 예시: NOR2X0_RVT pin 위치에서 위반, 원인은 ref library.

layout · imin_1/U123 (NOR2X0_RVT)



관찰

imin_1/U123 (NOR2X0_RVT) pin 위 VIA1 · M2 연결이 grid를 벗어남.

원인

OR / NOR gate cell의 ref library 문제.

조치

수정된 ref library로 교체 후 기존 flow 동일하게 재실행.

route_opt — check_routes

route_opt 후에도 DRC · open net 0 유지.

check_routes · after route_opt

PASS

```
Verify Summary:

Total number of nets = 312, of which 0 are not extracted
Total number of open nets = 0, of which 0 are frozen
Total number of excluded ports = 0 ports of 0 unplaced cells connected to 0 nets
                                 0 ports without pins of 0 cells connected to 0 nets
                                 0 ports of 0 cover cells connected to 0 non-pg nets
Total number of DRCs = 0
Total number of antenna violations = no antenna rules defined
Total number of tie to rail violations = not checked
Total number of tie to rail directly violations = not checked
```

0

DRCs

0

Open nets

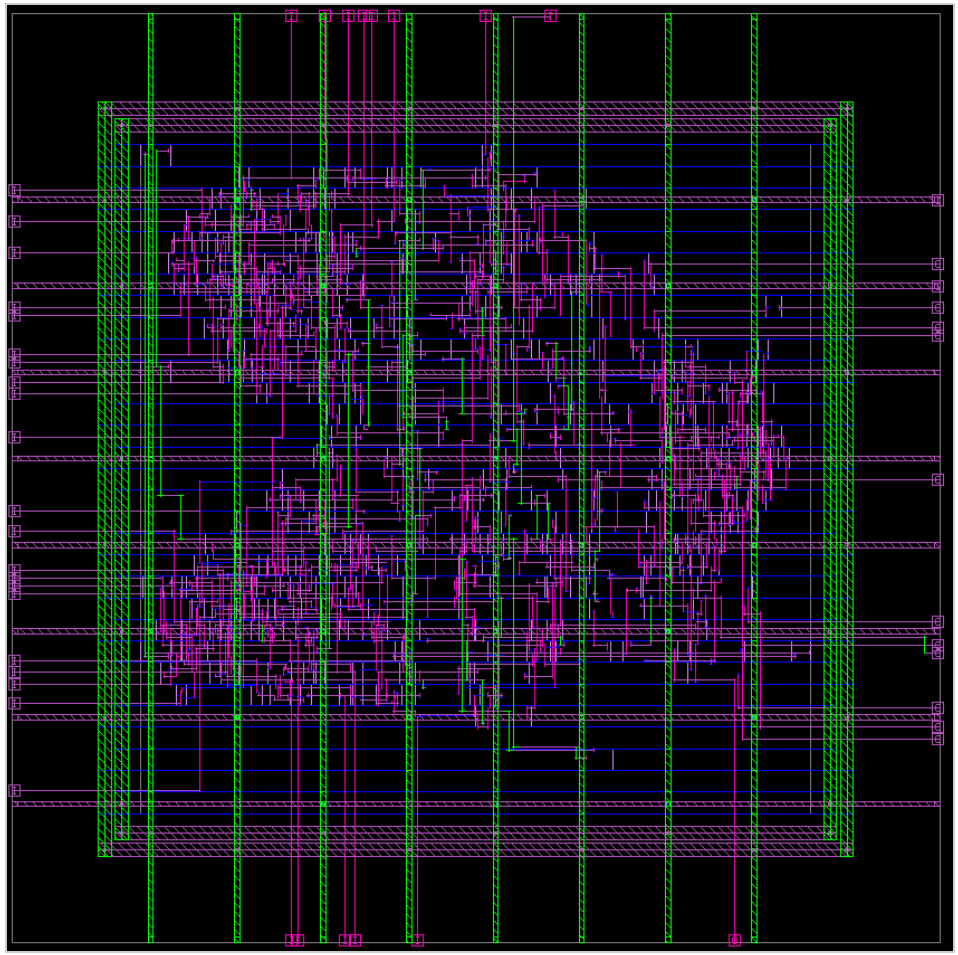
312

Nets

route_opt Layout

post-route 최적화 완료 layout.

layout · after route_opt



route_opt — Legality

check_legality — placement legality 통과.

check_legality · after route_opt

SUCCEEDED

```
check_legality for block design TOP succeeded!


check_legality succeeded.

**************************************

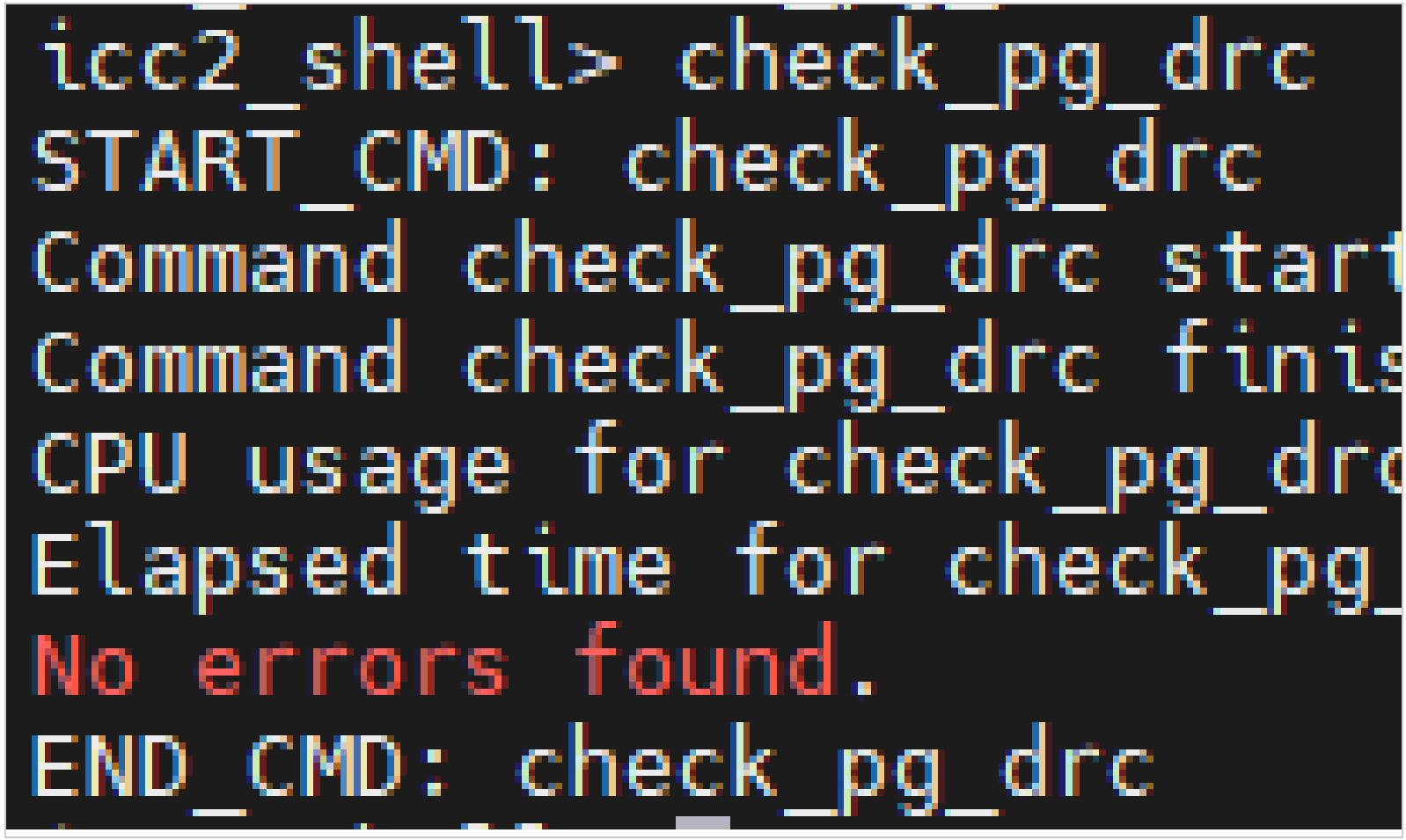
END_CMD: check_legality          CPU:      95 s (  0.
```

route_opt — PG DRC

check_pg_drc — No errors found.

check_pg_drc · after route_opt

PASS



route_opt — PG Connectivity

check_pg_connectivity — std cell 274개, floating 0.

check_pg_connectivity · after route_opt **PASS**

```
icc2_shell> check_pg_connectivity
START_CMD: check_pg_connectivity CPU:     86 s
Checking secondary net through power switch is
Secondary net will be checked together from pri
Primary Net : VDD    Secondary Net:
Primary Net : VSS    Secondary Net:
Loading cell instances...
Number of Standard Cells: 274
Number of Macro Cells: 0
Number of IO Pad Cells: 0
Number of Blocks: 0
Loading P/G wires and vias...
Number of VDD Wires: 28
Number of VDD Vias: 228
Number of VDD Terminals: 16
**************Verify net VDD connectivity******
  Number of floating wires: 0
  Number of floating vias: 0
  Number of floating std cells: 0
  Number of floating hard macros: 0
  Number of floating I/O pads: 0
  Number of floating terminals: 0
  Number of floating hierarchical blocks: 0
************************************************
Loading cell instances...
Loading P/G wires and vias...
Number of VSS Wires: 28
Number of VSS Vias: 228
Number of VSS Terminals: 16
**************Verify net VSS connectivity******
  Number of floating wires: 0
  Number of floating vias: 0
  Number of floating std cells: 0
  Number of floating hard macros: 0
  Number of floating I/O pads: 0
  Number of floating terminals: 0
  Number of floating hierarchical blocks: 0
************************************************
```

0

Floating

VDD · VSS

274

Std cells

06

SECTION

Filler & Sign-off

filler cell 삽입 후 DRC · LVS · PG · timing을 최종 점검.

정태웅 · 청주대학교

route.tcl (3/3)

filler cell 삽입 → legality 확인 → sign-off style check.

route.tcl · 6–7

```
#--------------------------------------------------------------------------
# 6. Filler cell insertion (after all ECO is finished)
#--------------------------------------------------------------------------
set FILLERS [get_lib_cells */*FILL*]
get_object_name $FILLERS

create_stdcell_fillers -lib_cells $FILLERS
connect_pg_net
remove_stdcell_fillers_with_violation

# If check_legality fails, remove the fillers and insert again:
#   remove_cells [get_cells -hierarchical -filter "ref_name=~*FILL*"]
check_legality
remove_cells [get_cells -hierarchical -filter "ref_name=~*FILL*"]
create_stdcell_fillers -lib_cells $FILLERS
connect_pg_net
remove_stdcell_fillers_with_violation

#--------------------------------------------------------------------------
# 7. Final sign-off style checks and save
#--------------------------------------------------------------------------
check_legality
check_routes
check_lvs -open_reporting detailed -check_child_cells true -max_errors 9999
check_pg_drc
check_pg_connectivity

report_utilization
report_threshold_voltage_groups
report_qor -summary -significant_digits 5
report_qor
report_constraints -all_violators -scenarios {normal_tt0p78v125c normal_tt1p05vn40c}
report_timing -delay_type max -scenarios {normal_tt0p78v125c normal_tt1p05vn40c} -report_by scenario
report_timing -delay_type min -scenarios {normal_tt0p78v125c normal_tt1p05vn40c} -report_by scenario
report_clock_qor

save_lib -as nlib/after_filler.nlib
```

Filler
*/*FILL* lib cell로 create_stdcell_fillers → connect_pg_net → 위반 filler 제거.

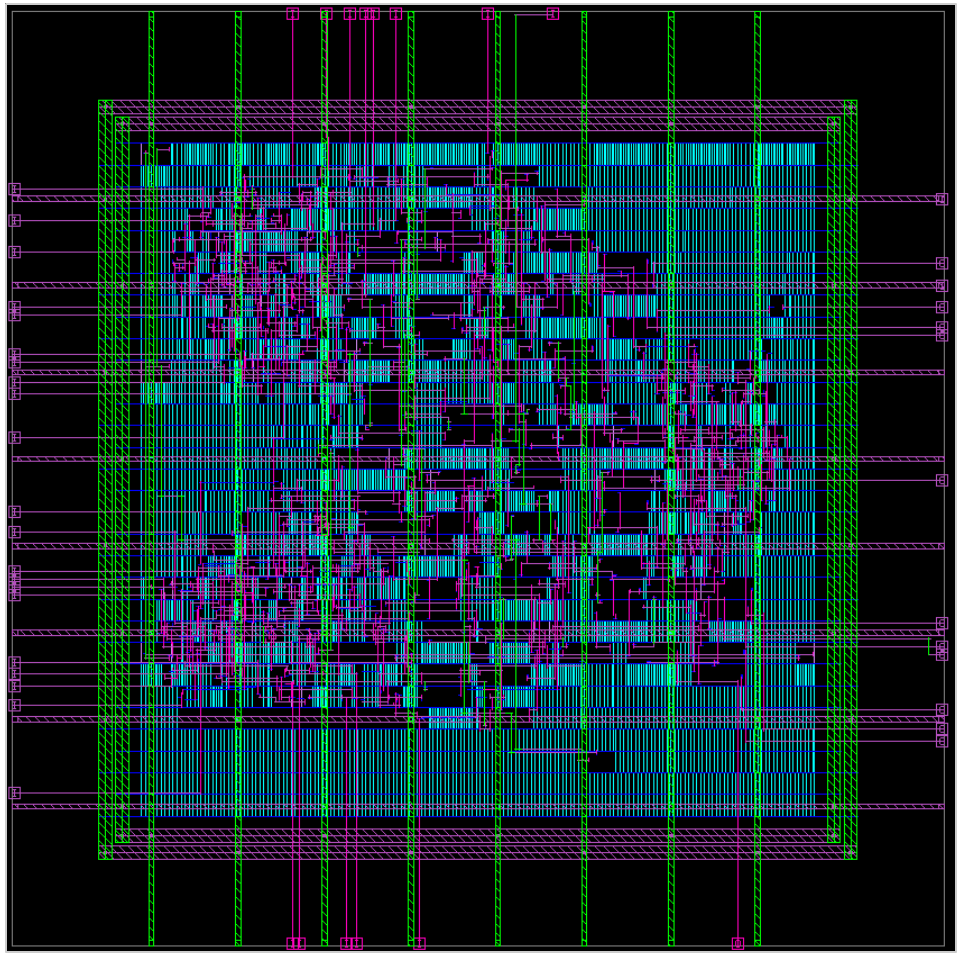
재삽입
legality 확인 후 filler 제거 · 재삽입.

Check
check_legality · check_routes · check_lvs · check_pg_drc · check_pg_connectivity.

Filler Insertion

빈 site를 filler cell로 채운 최종 layout.

layout · after filler



3,070

Filler cells

physical-only

99.89 %

Site 채움

2,691.39 / 2,694.43 μm²

check_routes — Final

filler 삽입 후 DRC 0 · open net 0.

check_routes · after filler

PASS

```
Verify Summary:

Total number of nets = 312, of which 0 are not extracted
Total number of open nets = 0, of which 0 are frozen
Total number of excluded ports = 0 ports of 0 unplaced cells connected to 0 nets
                                 0 ports without pins of 0 cells connected to 0 nets
                                 0 ports of 0 cover cells connected to 0 non-pg nets
Total number of DRCs = 0
Total number of antenna violations = no antenna rules defined
Total number of tie to rail violations = not checked
Total number of tie to rail directly violations = not checked
```

0
DRCs

0
Open nets

99.43 %
Redundant via
1,907 / 1,918

check_lvs

short · open · floating route 위반 0 (input net 312).

check_lvs · after filler

PASS

```
icc2_shell> check_lvs -open_reporting detailed -check_child_cells true -max_errors 9999
START_CMD: check_lvs           CPU:    102 s ( 0.03 hr) ELAPSE:    670 s ( 0.19 hr) MEM-
Information: Using 8 threads for LVS
[Check Net] Init         Elapsed =    0:00:00, CPU =    0:00:00
[Check Short] Stage 1    Elapsed =    0:00:00, CPU =    0:00:00
[Check Net] 10%          Elapsed =    0:00:00, CPU =    0:00:00
[Check Short] Stage 1-2 Elapsed =    0:00:00, CPU =    0:00:00
[Check Net] 20%          Elapsed =    0:00:00, CPU =    0:00:00
[Check Net] 30%          Elapsed =    0:00:00, CPU =    0:00:00
[Check Net] 40%          Elapsed =    0:00:00, CPU =    0:00:00
[Check Net] 50%          [Check Short] Stage 2   Elapsed =    0:00:00, CPU =    0:00:00
Elapsed =    0:00:00, CPU =    0:00:00
[Check Short] Stage 2-2 Elapsed =    0:00:00, CPU =    0:00:00
[Check Short] Stage 3    Elapsed =    0:00:00, CPU =    0:00:00
[Check Short] End        Elapsed =    0:00:00, CPU =    0:00:00
[Check Net] 60%          Elapsed =    0:00:00, CPU =    0:00:00
[Check Net] 70%          Elapsed =    0:00:00, CPU =    0:00:00
[Check Net] 80%          Elapsed =    0:00:00, CPU =    0:00:00
[Check Net] 90%          Elapsed =    0:00:00, CPU =    0:00:00
[Check Net] All nets are submitted.
[Check Net] 100%         Elapsed =    0:00:00, CPU =    0:00:00

************************************************************
    Maximum number of violations is set to 9999
    Abort checking when more than 9999 violations are found
    All violations might not be found.
************************************************************
Total number of input nets is 312.
Total number of short violations is 0.
Total number of open locations is 0 (0 open nets).
Total number of floating route violations is 0.
```

0

Short

0

Open

0

Floating route

check_legality — Final

filler 포함 placement legality 통과.

check_legality · after filler

SUCCEEDED

```
check_legality for block design TOP succeeded!


check_legality succeeded.

**************************************

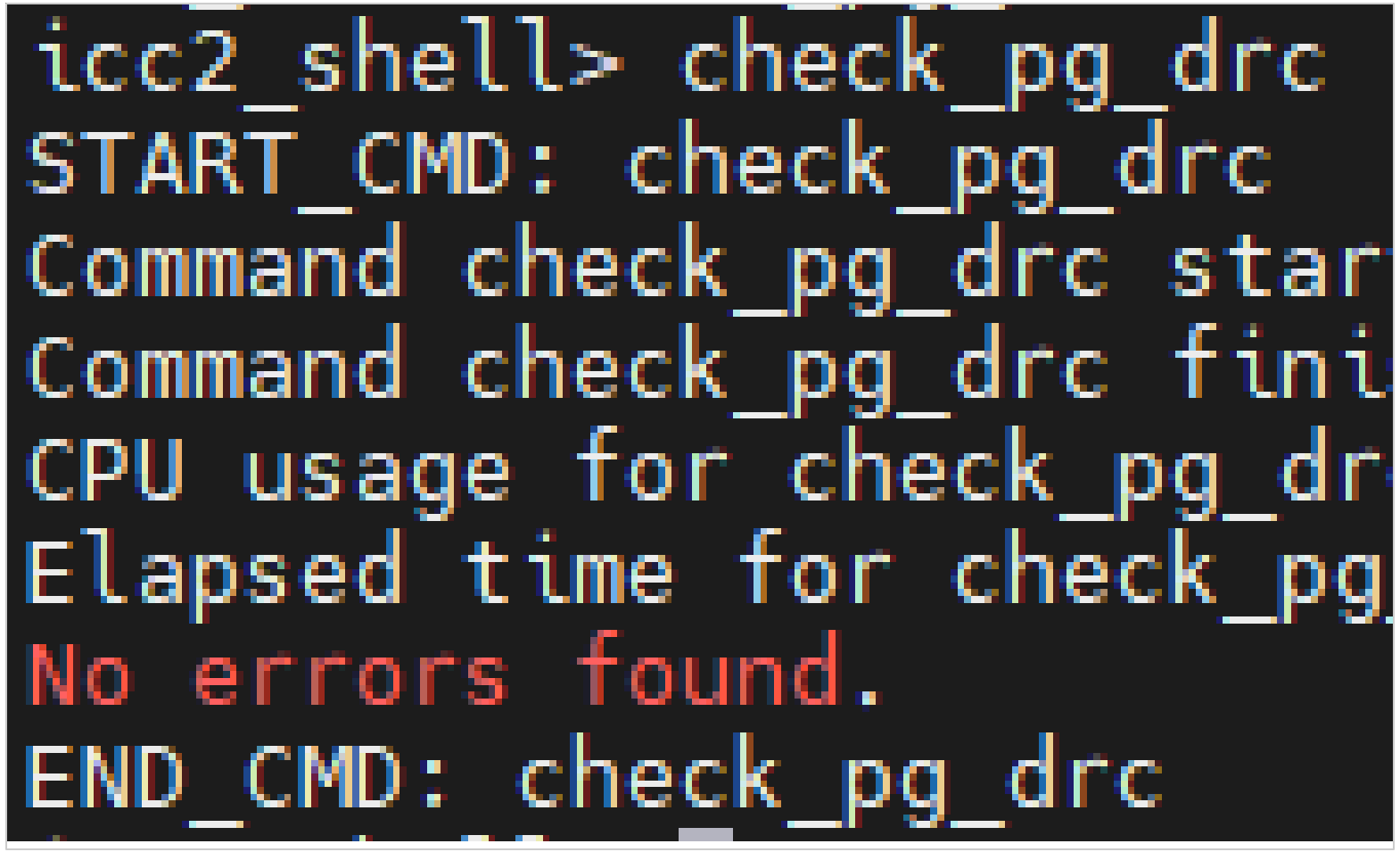
END_CMD: check_legality          CPU:    106 s (
```

PG DRC — Final

check_pg_drc — No errors found.

check_pg_drc · after filler

PASS



PG Connectivity — Final

filler 포함 std cell 3,344개, floating 0.

check_pg_connectivity · after filler — PASS

```
Number of VDD Wires: 28
Number of VDD Vias: 228
Number of VDD Terminals: 16
**************Verify net VDD connectivity*****************
  Number of floating wires: 0
  Number of floating vias: 0
  Number of floating std cells: 0
  Number of floating hard macros: 0
  Number of floating I/O pads: 0
  Number of floating terminals: 0
  Number of floating hierarchical blocks: 0
************************************************************
Loading cell instances...
Loading P/G wires and vias...
Number of VSS Wires: 28
Number of VSS Vias: 228
Number of VSS Terminals: 16
**************Verify net VSS connectivity*****************
  Number of floating wires: 0
  Number of floating vias: 0
  Number of floating std cells: 0
  Number of floating hard macros: 0
  Number of floating I/O pads: 0
  Number of floating terminals: 0
  Number of floating hierarchical blocks: 0
************************************************************
```

0

Floating

VDD · VSS

3,344

Std cells

filler 포함

QoR Summary — Final

hold 위반 0, tt0p78v125c setup −0.00031 ns 1 path 잔여.

report_qor -summary · after filler

```
icc2_shell> report_qor -summary -significant_digits 5
****************************************
Report : qor
         -summary
Design : TOP
Version: T-2022.03-SP5
Date   : Mon Oct 12 11:41:44 2026
****************************************
Information: Timer using 'SI, Timing Window Analysis, CRPR'. (TIM-050)

Timing
--------------------------------------------------------------------------------
Context                                 WNS             TNS             NVE
--------------------------------------------------------------------------------
normal_tt0p78v125c (Setup)          -0.00031        -0.00031               1
normal_tt1p05vn40c (Setup)           0.20554         0.00000               0
Design             (Setup)          -0.00031        -0.00031               1

normal_tt0p78v125c (Hold)            0.00686         0.00000               0
normal_tt1p05vn40c (Hold)            0.00802         0.00000               0
Design             (Hold)            0.00686         0.00000               0
--------------------------------------------------------------------------------

Miscellaneous
--------------------------------------------------------------------------------
Cell Area (netlist):                          888.48742
Cell Area (netlist and physical only):       2691.38500
Nets with DRC Violations:        0
```

−0.00031
Setup WNS (ns)
tt0p78v125c · 1 path

+0.00686
Hold WNS (ns)
NVE 0

잔여 setup
−0.31 ps 수준 — PrimeTime 단계에서 ECO로 처리 예정.

report_qor (1/3)

tt0p78v125c path group별 결과.

report_qor · tt0p78v125c

```
Scenario           'normal_tt0p78v125c'
Timing Path Group  '**in2reg_default**'
----------------------------------------
Levels of Logic:                     7
Critical Path Length:             0.82
Critical Path Slack:             -0.00
Critical Path Clk Period:         1.00
Total Negative Slack:            -0.00
No. of Violating Paths:              1
Worst Hold Violation:             0.00
Total Hold Violation:             0.00
No. of Hold Violations:              0
----------------------------------------

Scenario           'normal_tt0p78v125c'
Timing Path Group  '**reg2out_default**'
----------------------------------------
Levels of Logic:                     0
Critical Path Length:             0.22
Critical Path Slack:              0.07
Critical Path Clk Period:         1.00
Total Negative Slack:             0.00
No. of Violating Paths:              0
Worst Hold Violation:             0.00
Total Hold Violation:             0.00
No. of Hold Violations:              0
----------------------------------------

Scenario           'normal_tt0p78v125c'
Timing Path Group  'clk'
----------------------------------------
Levels of Logic:                     6
Critical Path Length:             0.89
Critical Path Slack:              0.01
Critical Path Clk Period:         1.00
Total Negative Slack:             0.00
No. of Violating Paths:              0
Worst Hold Violation:             0.00
Total Hold Violation:             0.00
No. of Hold Violations:              0
----------------------------------------
```

in2reg
logic 7 level, slack −0.00, setup 위반 1.

reg2out
slack 0.07, 위반 0.

clk
logic 6 level, slack 0.01, 위반 0.

report_qor (2/3)

tt1p05vn40c path group별 결과 — 위반 0.

report_qor · tt1p05vn40c

```
Scenario           'normal_tt1p05vn40c'
Timing Path Group  '**in2reg_default**'
----------------------------------------
Levels of Logic:                     9
Critical Path Length:             0.51
Critical Path Slack:              0.38
Critical Path Clk Period:         1.00
Total Negative Slack:             0.00
No. of Violating Paths:              0
Worst Hold Violation:             0.00
Total Hold Violation:             0.00
No. of Hold Violations:              0
----------------------------------------

Scenario           'normal_tt1p05vn40c'
Timing Path Group  '**reg2out_default**'
----------------------------------------
Levels of Logic:                     0
Critical Path Length:             0.12
Critical Path Slack:              0.21
Critical Path Clk Period:         1.00
Total Negative Slack:             0.00
No. of Violating Paths:              0
Worst Hold Violation:             0.00
Total Hold Violation:             0.00
No. of Hold Violations:              0
----------------------------------------

Scenario           'normal_tt1p05vn40c'
Timing Path Group  'clk'
----------------------------------------
Levels of Logic:                     8
Critical Path Length:             0.45
Critical Path Slack:              0.49
Critical Path Clk Period:         1.00
Total Negative Slack:             0.00
No. of Violating Paths:              0
Worst Hold Violation:             0.00
Total Hold Violation:             0.00
No. of Hold Violations:              0
----------------------------------------
```

in2reg
logic 9 level, slack 0.38, 위반 0.

reg2out
slack 0.21, 위반 0.

clk
logic 8 level, slack 0.49, 위반 0.

report_qor (3/3)

cell count · area · design rule.

report_qor · cell count · area · design rules **0 DRC**

```
Cell Count
----------------------------------------
Hierarchical Cell Count:              3
Hierarchical Port Count:             74
Leaf Cell Count:                    274
Buf/Inv Cell Count:                 108
Buf Cell Count:                      42
Inv Cell Count:                      66
CT Buf/Inv Cell Count:                0
Combinational Cell Count:           214
   Single-bit Isolation Cell Count:                    0
   Multi-bit Isolation Cell Count:                     0
   Isolation Cell Banking Ratio:                       0.00%
   Single-bit Level Shifter Cell Count:                0
   Multi-bit Level Shifter Cell Count:                 0
   Level Shifter Cell Banking Ratio:                   0.00%
   Single-bit ELS Cell Count:                          0
   Multi-bit ELS Cell Count:                           0
   ELS Cell Banking Ratio:                             0.00%
Sequential Cell Count:               60
   Integrated Clock-Gating Cell Count:                 0
   Sequential Macro Cell Count:                        0
   Single-bit Sequential Cell Count:                   60
   Multi-bit Sequential Cell Count:                    0
   Sequential Cell Banking Ratio:                      0.00%
   BitsPerflop:                                        1.00
Macro Count:                         0
----------------------------------------


Area
----------------------------------------
Combinational Area:             461.53
Noncombinational Area:          426.96
Buf/Inv Area:                   197.98
Total Buffer Area:              113.60
Total Inverter Area:             84.38
Macro/Black Box Area:             0.00
Net Area:                            0
Net XLength:                  2776.52
Net YLength:                  2580.01
----------------------------------------
Cell Area (netlist):                          888.49
Cell Area (netlist and physical only):       2691.39
Net Length:                   5356.53


Design Rules
----------------------------------------
Total Number of Nets:           312
Nets with Violations:             0
Max Trans Violations:             0
Max Cap Violations:               0
----------------------------------------
```

274

Leaf cells

comb 214 · seq 60

888.49

Cell area (μm²)

netlist

0

DRC violations

312 nets

Buf / Inv

108 (buf 42 · inv 66).

Setup Timing — tt0p78v125c

imin2[0] → imin_1/clk_r_REG34_S1 (in2reg) — 잔여 setup 1 path.

report_timing -delay_type max · tt0p78v125c **VIOLATED**

```
Startpoint: imin2[0] (input port clocked by clk)
Endpoint: imin_1/clk_r_REG34_S1 (rising edge-triggered flip-flop clocked by clk)
Mode: normal
Corner: tt0p78v125c
Scenario: normal_tt0p78v125c
Path Group: **in2reg_default**
Path Type: max

Point                                          Incr       Path
-----------------------------------------------------------------
clock clk (rise edge)                          0.00       0.00
clock network delay (propagated)               0.80       0.82
input external delay                           0.18       1.00 r
imin2[0] (in)                                  0.01       1.00 r
imin_1/U107/Y (INVX0_HVT)                      0.08       1.08 f
imin_1/ctmTdsLR_2_401/Y (OA222X1_LVT)          0.09       1.17 f
imin_1/ctmTdsLR_2_429/Y (AO21X1_RVT)           0.09       1.25 f
imin_1/ctmTdsLR_1_428/Y (OA21X1_RVT)           0.08       1.33 f
imin_1/U33/Y (OA221X1_RVT)                     0.14       1.47 f
imin_1/U34/Y (AND2X1_RVT)                      0.09       1.55 f
imin_1/U52/Y (AO22X1_RVT)                      0.07       1.62 f
imin_1/clk_r_REG34_S1/D (DFFARX1_RVT)          0.00       1.62 f
data arrival time                                         1.62

clock clk (rise edge)                          1.00       1.00
clock network delay (propagated)               0.68       1.68
clock reconvergence pessimism                  0.00       1.68
imin_1/clk_r_REG34_S1/CLK (DFFARX1_RVT)        0.00       1.68 r
clock uncertainty                             -0.02       1.66
library setup time                            -0.04       1.62
data required time                                        1.62
-----------------------------------------------------------------
data required time                                        1.62
data arrival time                                        -1.62
-----------------------------------------------------------------
slack (VIOLATED)                                         -0.00
```

1.62
Arrival (ns)

1.62
Required (ns)

−0.00031
Slack (ns)
qor summary 기준

판단
−0.31 ps — PrimeTime 단계에서 ECO 예정.

Setup Timing — tt1p05vn40c

clk_r_REG5_S4 → error[2] (reg2out), slack 0.21.

report_timing -delay_type max · tt1p05vn40c

MET

```
Startpoint: clk_r_REG5_S4 (rising edge-triggered flip-flop clocked by clk)
Endpoint: error[2] (output port clocked by clk)
Mode: normal
Corner: tt1p05vn40c
Scenario: normal_tt1p05vn40c
Path Group: **reg2out_default**
Path Type: max

Point                                              Incr      Path
-----------------------------------------------------------------
clock clk (rise edge)                              0.00      0.00
clock network delay (propagated)                   0.50      0.50

clk_r_REG5_S4/CLK (DFFARX1_RVT)                    0.00      0.50 r
clk_r_REG5_S4/Q (DFFARX1_RVT)                      0.12      0.62 r
error[2] (out)                                     0.00      0.62 r
data arrival time                                            0.62

clock clk (rise edge)                              1.00      1.00
clock network delay (propagated)                   0.45      1.45
clock reconvergence pessimism                      0.00      1.45
clock uncertainty                                 -0.02      1.43
output external delay                             -0.61      0.82
data required time                                           0.82
-----------------------------------------------------------------
data required time                                           0.82
data arrival time                                           -0.62
-----------------------------------------------------------------
slack (MET)                                                  0.21
```

0.62

Arrival (ns)

0.82

Required (ns)

0.21

Slack (ns)

Hold Timing — tt0p78v125c

imax2[0] → imax_1/clk_r_REG24_S1_IP, slack +0.00686.

report_timing -delay_type min · tt0p78v125c

MET

```
Startpoint: imax2[0] (input port clocked by clk)
Endpoint: imax_1/clk_r_REG24_S1_IP (rising edge-triggered flip-flop clocked by clk)
Mode: normal
Corner: tt0p78v125c
Scenario: normal_tt0p78v125c
Path Group: **in2reg_default**
Path Type: min

Point                                                  Incr       Path
-----------------------------------------------------------------------
clock clk (rise edge)                                  0.00       0.00
clock network delay (propagated)                       0.71       0.72
input external delay                                   0.05       0.77 f
imax2[0] (in)                                          0.01       0.77 f
imax_1/U88/Y (INVX0_HVT)                               0.04       0.81 r
imax_1/clk_r_REG24_S1_IP/D (DFFASX1_RVT)              -0.01       0.80 r
data arrival time                                                 0.80

clock clk (rise edge)                                  0.00       0.00
clock network delay (propagated)                       0.79       0.79
clock reconvergence pessimism                         -0.00       0.79
imax_1/clk_r_REG24_S1_IP/CLK (DFFASX1_RVT)             0.00       0.79 r
clock uncertainty                                      0.03       0.82
library hold time                                     -0.03       0.80
data required time                                                0.80
-----------------------------------------------------------------------
data required time                                                0.80
data arrival time                                                -0.80
-----------------------------------------------------------------------
slack (MET)                                                       0.01
```

0.80

Arrival (ns)

0.80

Required (ns)

+0.00686

Slack (ns)

qor summary 기준

Hold Timing — tt1p05vn40c

imax4[2] → imax_1/clk_r_REG29_S1_IP, slack +0.00802.

report_timing -delay_type min · tt1p05vn40c

MET

```
Startpoint: imax4[2] (input port clocked by clk)
Endpoint: imax_1/clk_r_REG29_S1_IP (rising edge-triggered flip-flop clocked by clk)
Mode: normal
Corner: tt1p05vn40c
Scenario: normal_tt1p05vn40c
Path Group: **in2reg_default**
Path Type: min

Point                                              Incr       Path
-----------------------------------------------------------------------
clock clk (rise edge)                              0.00       0.00
clock network delay (propagated)                   0.45       0.46
input external delay                               0.05       0.51 r
imax4[2] (in)                                      0.01       0.51 r
imax_1/U55/Y (INVX0_HVT)                           0.02       0.54 f
imax_1/clk_r_REG29_S1_IP/D (DFFASX1_RVT)           0.00       0.54 f
data arrival time                                             0.54

clock clk (rise edge)                              0.00       0.00
clock network delay (propagated)                   0.50       0.50
clock reconvergence pessimism                     -0.00       0.50
imax_1/clk_r_REG29_S1_IP/CLK (DFFASX1_RVT)         0.00       0.50 r
clock uncertainty                                  0.03       0.53
library hold time                                 -0.01       0.53
data required time                                            0.53
-----------------------------------------------------------------------
data required time                                            0.53
data arrival time                                            -0.54
-----------------------------------------------------------------------
slack (MET)                                                   0.01
```

0.54

Arrival (ns)

0.53

Required (ns)

+0.00802

Slack (ns)

qor summary 기준

Constraints — Final

design rule 위반 0 — late timing endpoint 1개 (imin_1/clk_r_REG34_S1/D, 잔여 setup).

report_constraints -all_violators · after filler

```
Mode: normal Corner: tt0p78v125c
Scenario: normal_tt0p78v125c
max_transition
                          Required       Actual
Net                      Transition    Transition       Slack  Violation
-------------------------------------------------------------------------
HFSNET_13                   0.12          0.12          0.00  (MET)

-------------------------------------------------------------------------
Number of max_transition violation(s): 0

Mode: normal Corner: tt1p05vn40c
Scenario: normal_tt1p05vn40c
max_transition
                          Required       Actual
Net                      Transition    Transition       Slack  Violation
-------------------------------------------------------------------------
imax_1/n34                  0.05          0.03          0.02  (MET)

-------------------------------------------------------------------------
Number of max_transition violation(s): 0

Mode: normal Corner: tt0p78v125c
Scenario: normal_tt0p78v125c
max_capacitance
                          Required       Actual
Net                      Capacitance   Capacitance      Slack  Violation
-------------------------------------------------------------------------
HFSNET_9                    10.00          9.99          0.01  (MET)

-------------------------------------------------------------------------
Number of max_capacitance violation(s): 0

Mode: normal Corner: tt1p05vn40c
Scenario: normal_tt1p05vn40c
max_capacitance
                          Required       Actual
Net                      Capacitance   Capacitance      Slack  Violation
-------------------------------------------------------------------------
HFSNET_9                    10.00          9.74          0.26  (MET)

-------------------------------------------------------------------------
Number of max_capacitance violation(s): 0


Mode: normal Corner: tt0p78v125c
Scenario: normal_tt0p78v125c
-------------------------------------------------------------------------
Number of min_capacitance violation(s): 0

Mode: normal Corner: tt1p05vn40c
Scenario: normal_tt1p05vn40c
-------------------------------------------------------------------------
Number of min_capacitance violation(s): 0

Total number of violation(s): 0
```

0

DRC

trans · cap

1

Setup endpoint

tt0p78v125c

Clock QoR — Final

route 후 clock latency · skew, trans · cap DRC 0.

report_clock_qor · after filler

```
============================================================ Summary Table for Corner tt0p78v125c ============================================================
Clock /                                 Attrs     Sinks Levels    Clock     Clock     Clock       Max    Global  Trans DRC  Cap DRC      Wire
Skew Group                                                       Repeater  Repeater  Stdcell   Latency      Skew     Count     Count    Length
                                                                    Count      Area      Area
------------------------------------------------------------------------------------------------------------------------------------------------
### Mode: normal, Scenario: normal_tt0p78v125c
clk                                      M,D        60     13       13     27.96     69.89      0.62      0.03         0         0    652.24
------------------------------------------------------------------------------------------------------------------------------------------------
All Clocks                                          60     13       13     27.96     69.89      0.62      0.03         0         0    652.24


Warning: Please use -largest / -smallest / -all switches with -show_verbose_paths / -show_paths to report the clock paths. (CTS-956)
================================================
==== Summary Reporting for Corner tt1p05vn40c ====
================================================

============================================================ Summary Table for Corner tt1p05vn40c ============================================================
Clock /                                 Attrs     Sinks Levels    Clock     Clock     Clock       Max    Global  Trans DRC  Cap DRC      Wire
Skew Group                                                       Repeater  Repeater  Stdcell   Latency      Skew     Count     Count    Length
                                                                    Count      Area      Area
------------------------------------------------------------------------------------------------------------------------------------------------
### Mode: normal, Scenario: normal_tt1p05vn40c
clk                                      M,D        60     13       13     27.96     69.89      0.34      0.02         0         0    652.24
------------------------------------------------------------------------------------------------------------------------------------------------
All Clocks                                          60     13       13     27.96     69.89      0.34      0.02         0         0    652.24
```

0.62 / 0.03

tt0p78v125c

latency / skew

0.34 / 0.02

tt1p05vn40c

latency / skew

0 / 0

Trans / Cap DRC

Vth Ratio — Final

표기상 LVT 91.93 %는 filler 포함 — logic cell만 보면 1.46 %.

report_threshold_voltage_group · after filler

```
icc2_shell> report_threshold_voltage_groups
****************************************
Report : threshold_voltage_group
Design : TOP
Version: T-2022.03-SP5
Date   : Mon Oct 12 11:42:18 2026
****************************************

Cell Count Report
-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------

Group Names                     All (%)       Repeaters (%)   Combinational (%)       Registers (%)     Sequentials (%)   Clock_network (%)          Macro (%)   Physical_only (%)
-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------
saed32cell_hvt             91   (2.72%)        69   (2.06%)        19   (0.57%)         3   (0.09%)         0   (0.00%)         0   (0.00%)         0   (0.00%)         0   (0.00%)
saed32cell_lvt           3074  (91.93%)         0   (0.00%)         4   (0.12%)         0   (0.00%)         0   (0.00%)         0   (0.00%)         0   (0.00%)      3070  (91.81%)  L
saed32cell_svt            179   (5.35%)        16   (0.48%)        83   (2.48%)        57   (1.70%)         0   (0.00%)        23   (0.69%)         0   (0.00%)         0   (0.00%)
-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------
Total                    3344 (100.00%)        85   (2.54%)       106   (3.17%)        60   (1.79%)         0   (0.00%)        23   (0.69%)         0   (0.00%)      3070  (91.81%)
Low vth groups           3074  (91.93%)         0   (0.00%)         4   (0.12%)         0   (0.00%)         0   (0.00%)         0   (0.00%)         0   (0.00%)      3070  (91.81%)
-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------


Cell Area Report
-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------

Group Names                     All (%)       Repeaters (%)   Combinational (%)       Registers (%)     Sequentials (%)   Clock_network (%)          Macro (%)   Physical_only (%)
-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------
saed32cell_hvt          175.11   (6.51%)     107.50   (3.99%)     46.25   (1.72%)     21.35   (0.79%)      0.00   (0.00%)      0.00   (0.00%)      0.00   (0.00%)      0.00   (0.00%)
saed32cell_lvt         1813.57  (67.38%)       0.00   (0.00%)     10.67   (0.40%)      0.00   (0.00%)      0.00   (0.00%)      0.00   (0.00%)      0.00   (0.00%)   1802.90  (66.99%)  L
saed32cell_svt          702.71  (26.11%)      20.59   (0.76%)    206.62   (7.68%)    405.61  (15.07%)      0.00   (0.00%)     69.89   (2.60%)      0.00   (0.00%)      0.00   (0.00%)
-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------
Total                  2691.38 (100.00%)     128.09   (4.76%)    263.55   (9.79%)    426.96  (15.86%)      0.00   (0.00%)     69.89   (2.60%)      0.00   (0.00%)   1802.90  (66.99%)
Low vth groups         1813.57  (67.38%)       0.00   (0.00%)     10.67   (0.40%)      0.00   (0.00%)      0.00   (0.00%)      0.00   (0.00%)      0.00   (0.00%)   1802.90  (66.99%)
-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------
```

91.93 %

LVT (filler 포함)

3,074 / 3,344

1.46 %

LVT (logic)

4 / 274 · spec < 20 %

이유

physical-only filler 3,070개가 모두 LVT group.

Utilization — Final

report_utilization — 0.3297.

report_utilization · after filler

```
icc2_shell> report_utilization
****************************************
Report : report_utilization
Design : TOP
Version: T-2022.03-SP5
Date   : Mon Oct 12 11:42:58 2026
****************************************
Utilization Ratio:                      0.3297
Utilization options:
 - Area calculation based on:           site_row of block TOP
 - Categories of objects excluded:      hard_macros macro_keepouts soft_macros io_cells hard_blockages
Total Area:                             2694.4347
Total Capacity Area:                    2694.4347
Total Area of cells:                    888.4874
Area of excluded objects:
 - hard_macros          :               0.0000
 - macro_keepouts       :               0.0000
 - soft_macros          :               0.0000
 - io_cells             :               0.0000
 - hard_blockages       :               0.0000

Utilization of site-rows with:
 - Site 'unit':                         0.3297

0.3297
```

0.3297
Utilization

888.49
Cell area (μm²)

추이
floorplan 0.4488 → place 0.3077 → CTS 0.3285 → final 0.3297.

Stage-by-Stage QoR

Floorplan → Place → CTS → Route + Filler 단계별 scenario별 timing과 area · utilization 변화.

단계	Setup WNS (ns)		Hold WNS (ns)		Cell area (μm²)	Utilization
	tt0p78v125c	tt1p05vn40c	tt0p78v125c	tt1p05vn40c		
Floorplan (DC netlist)	–	–	–	–	1,209.22	0.4488
Place (refine_opt)	+0.01828	+0.20474	+0.03001	−0.00037 (1)	829.02	0.3077
CTS	+0.00972	+0.20892	+0.00121	+0.01039	885.18	0.3285
Route + Filler	−0.00031 (1)	**+0.20554**	**+0.00686**	**+0.00802**	888.49	0.3297

각 단계 report_qor -summary · report_utilization 기준. Floorplan 단계는 timing 리포트 없음, 면적은 coarse placement 시점 DC netlist cell area. (n) = 위반 path 수.

Final Results

Route + Filler 이후 최종 결과 — hold · DRC · LVS · PG 통과, setup 1 path(−0.31 ps)만 PrimeTime ECO 예정.

항목	tt0p78v125c	tt1p05vn40c
Setup WNS / TNS / NVE	−0.00031 / −0.00031 / 1	+0.20554 / 0 / 0
Hold WNS / TNS / NVE	+0.00686 / 0 / 0	+0.00802 / 0 / 0
Clock latency / skew	0.62 / 0.03	0.34 / 0.02
Max trans / Max cap 위반	0 / 0 (312 nets)	
check_routes DRC / open net	0 / 0	
LVS short / open / floating	0 / 0 / 0	
PG floating (VDD · VSS)	0	
Leaf cell (comb / seq)	274 (214 / 60)	
Buf / Inv cell	108 (42 / 66)	
Filler cell	3,070	
Cell area (netlist / +physical)	888.49 / 2,691.39 μm²	
Utilization	0.3297	
LVT (logic cell)	1.46 % (4 / 274, spec < 20 %)	
Redundant via / wire length	99.43 % / 3,851 μm	

−0.00031 ns

Setup WNS

1 path · PT ECO 예정

0 / 0

DRC / LVS

check_routes · check_lvs

1.46 %

LVT (logic)

spec < 20 %

+0.00686 ns

Hold WNS

tt0p78v125c · NVE 0

총정리

spec flow대로 setup부터 sign-off check까지 완주 — 이후 StarRC → PrimeTime → Second-pass Synthesis.

Flow
setup → floorplan · PNS → place → CTS → route → filler, place · CTS · route 진입 전 check_design error 0.

Issue ① TCK-012
tt1p05vn40c I/O constraint 미적용 — scenario별 재source로 34 → 2 (reset만 남음, 의도됨).

Issue ② zroute
route 단계 zroute error 21건 (OR / NOR gate) — ref library 문제로 확인, 수정된 ref로 재실행해 DRC 0.

Sign-off
check_routes DRC 0 · LVS short / open 0 · PG floating 0 · hold WNS +0.00686 ns · logic LVT 1.46 %, setup −0.31 ps 1 path는 PT ECO.

Next

StarRC

routed design에서 SPEF 추출.

↓

PrimeTime

SPEF 기반 cell + net delay STA, SDF 추출, 잔여 setup ECO.

↓

Second-pass Synthesis

floorplan DEF를 DC topographical에 반영해 재합성.